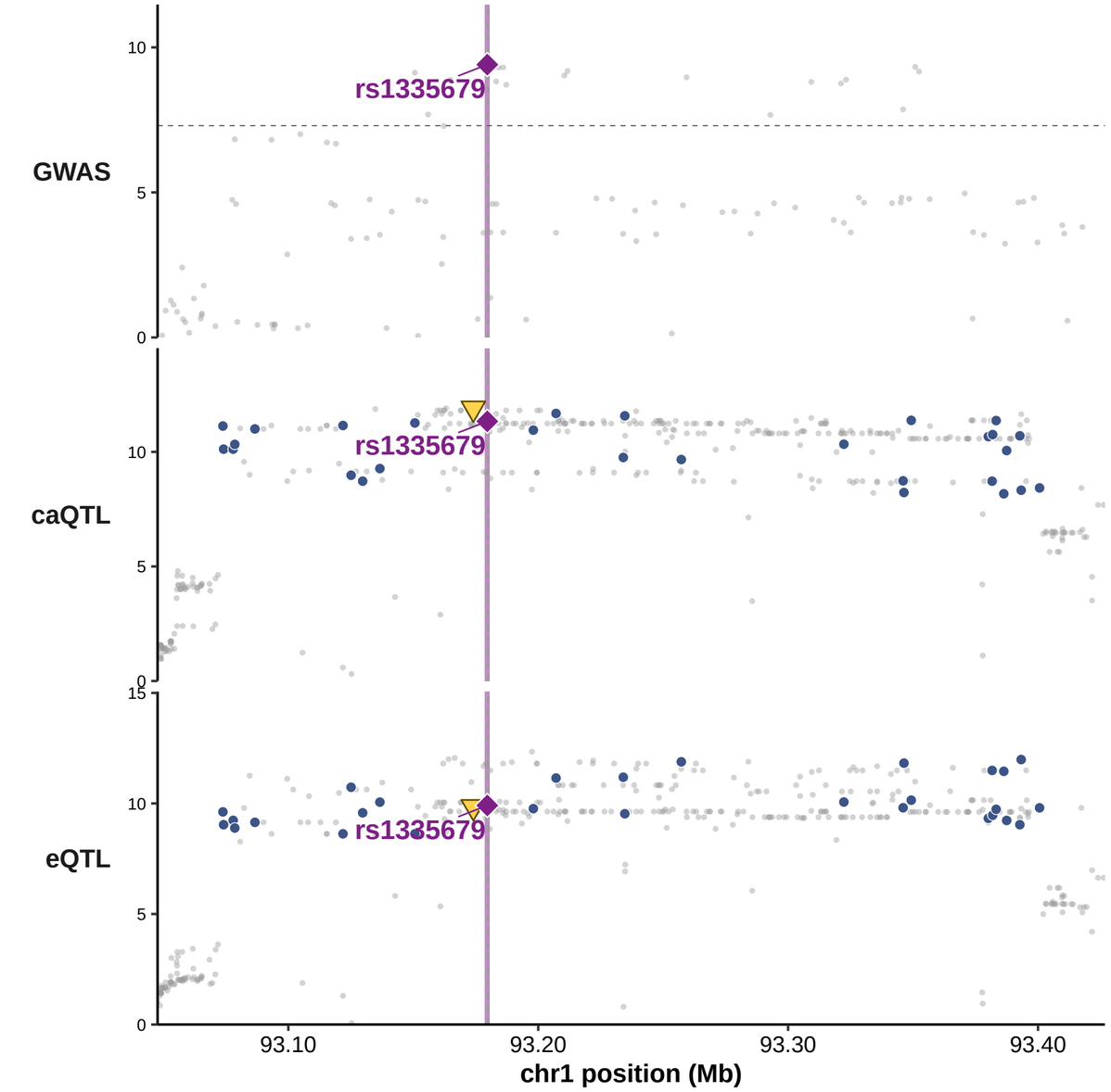
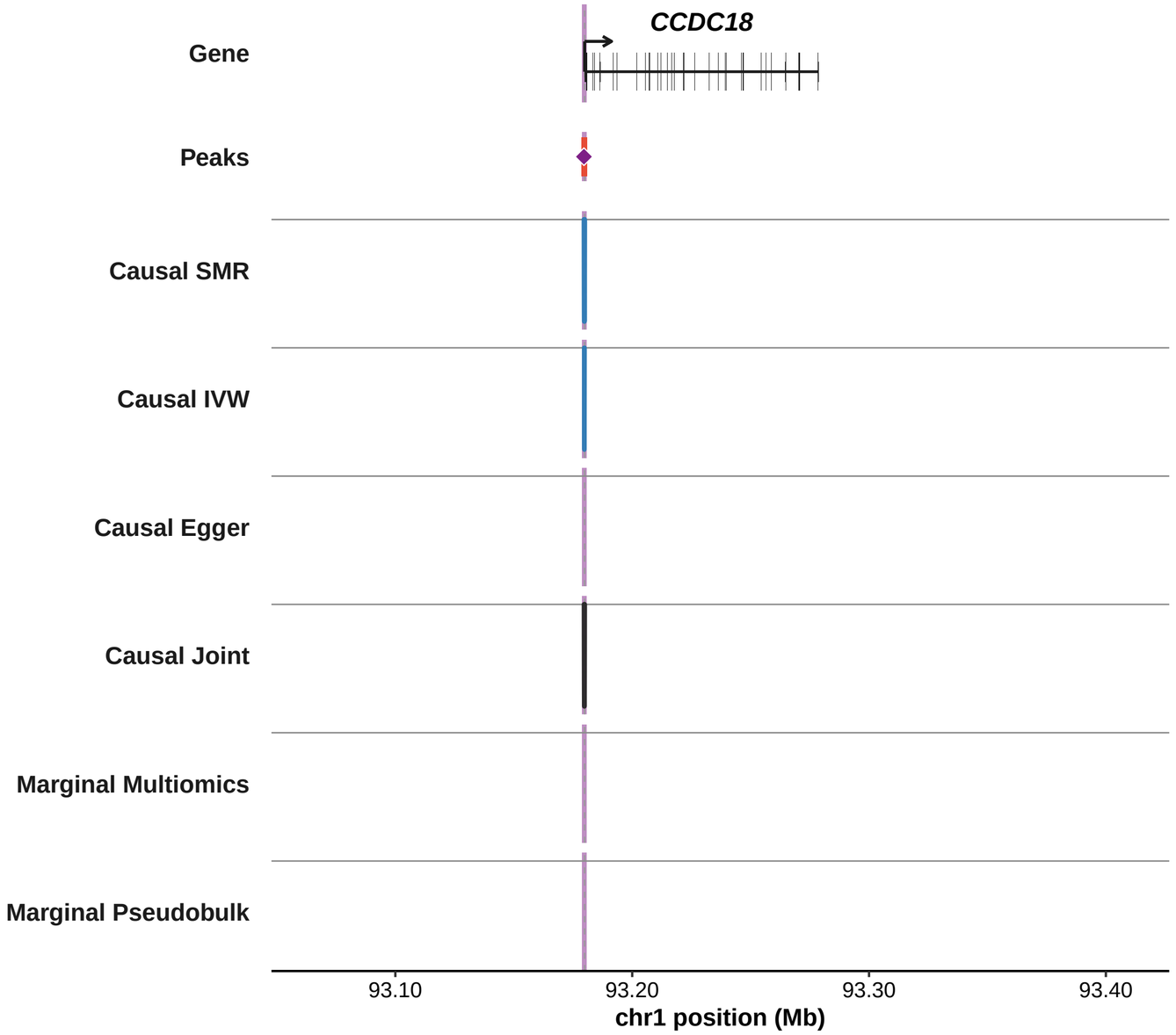
Bn_TCL1A − CCDC18 − chr1:93179440−93179941 − CD

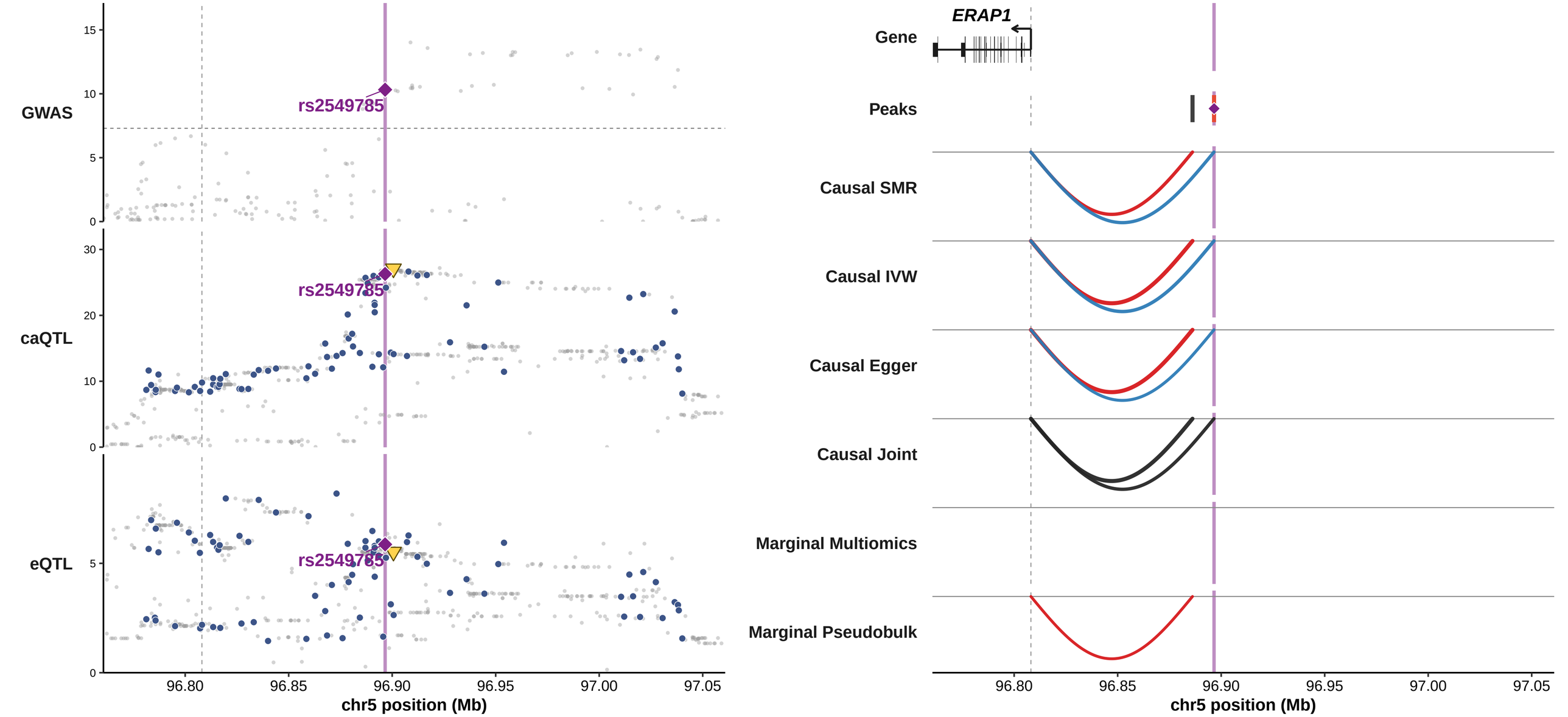
P−value: Causal SMR 1.9e−06 | Causal IVW 1.4e−04 | Causal Egger 9.8e−01 | Causal Joint 5.8e−06 | Marginal Multiomics 3.4e−01 | Marginal Pseudobulk 2.9e−02





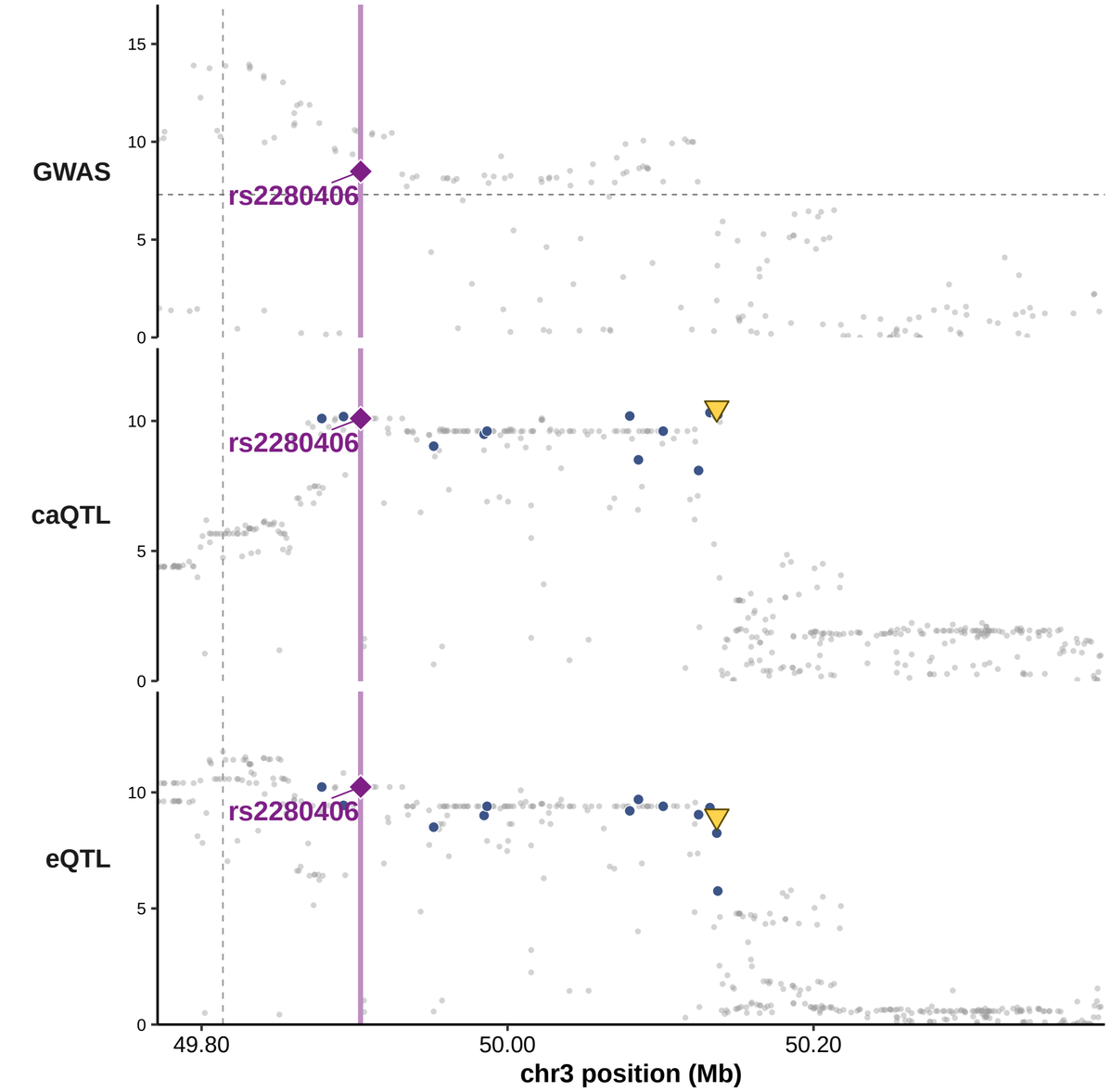
Bn_TCL1A − ERAP1 − chr5:96896259−96896760 − CD

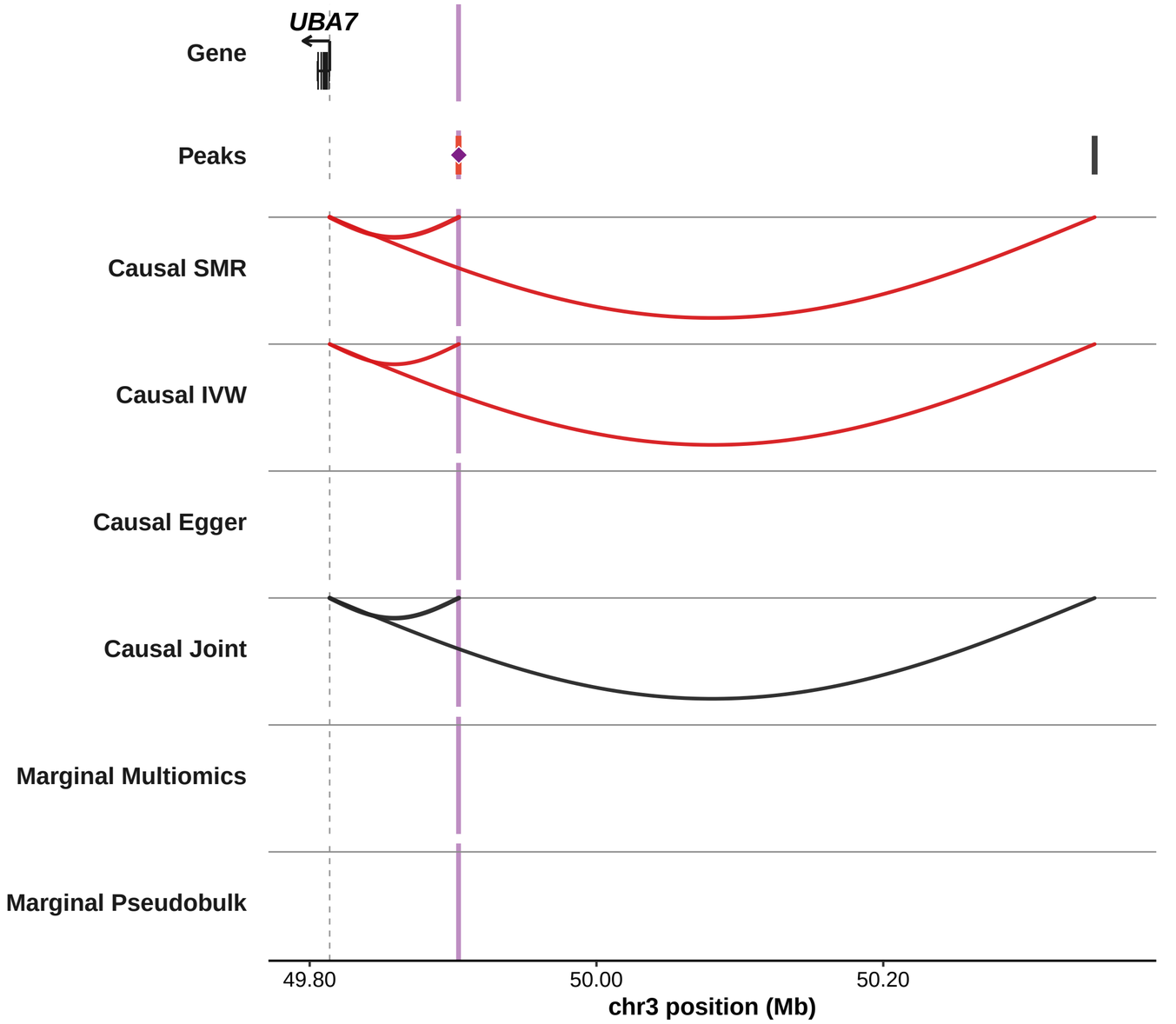
P−value: Causal SMR 1.7e−05 | Causal IVW 5.4e−06 | Causal Egger 3.2e−04 | Causal Joint 1.2e−05 | Marginal Multiomics 6.3e−02 | Marginal Pseudobulk 9.6e−01



Bn_TCL1A − UBA7 − chr3:49903714−49904215 − IBD

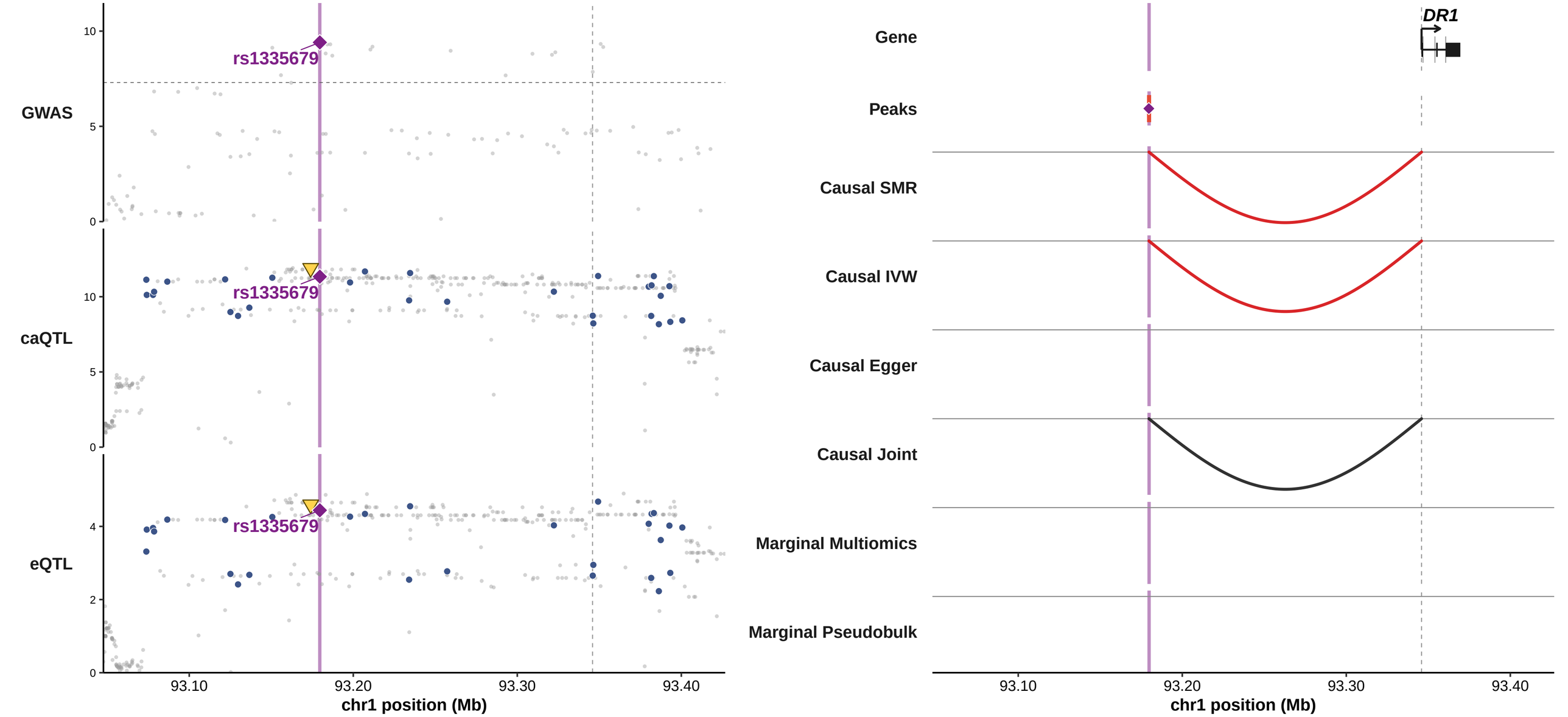
P−value: Causal SMR 7.1e−06 | Causal IVW 2.9e−03 | Causal Egger 5.5e−01 | Causal Joint 2.1e−05 | Marginal Multiomics 9.7e−01 | Marginal Pseudobulk 7.0e−02





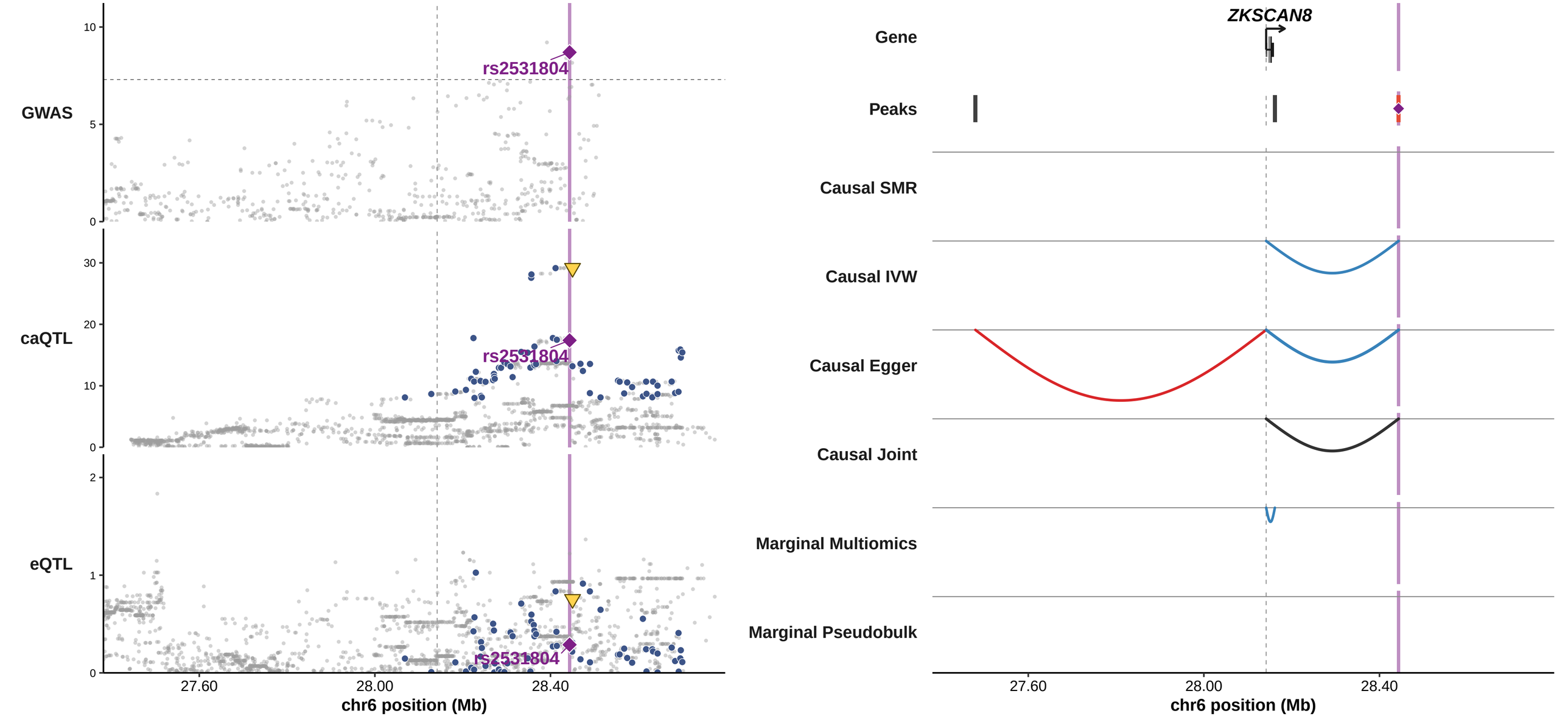
Bn_TCL1A − DR1 − chr1:93179440−93179941 − CD

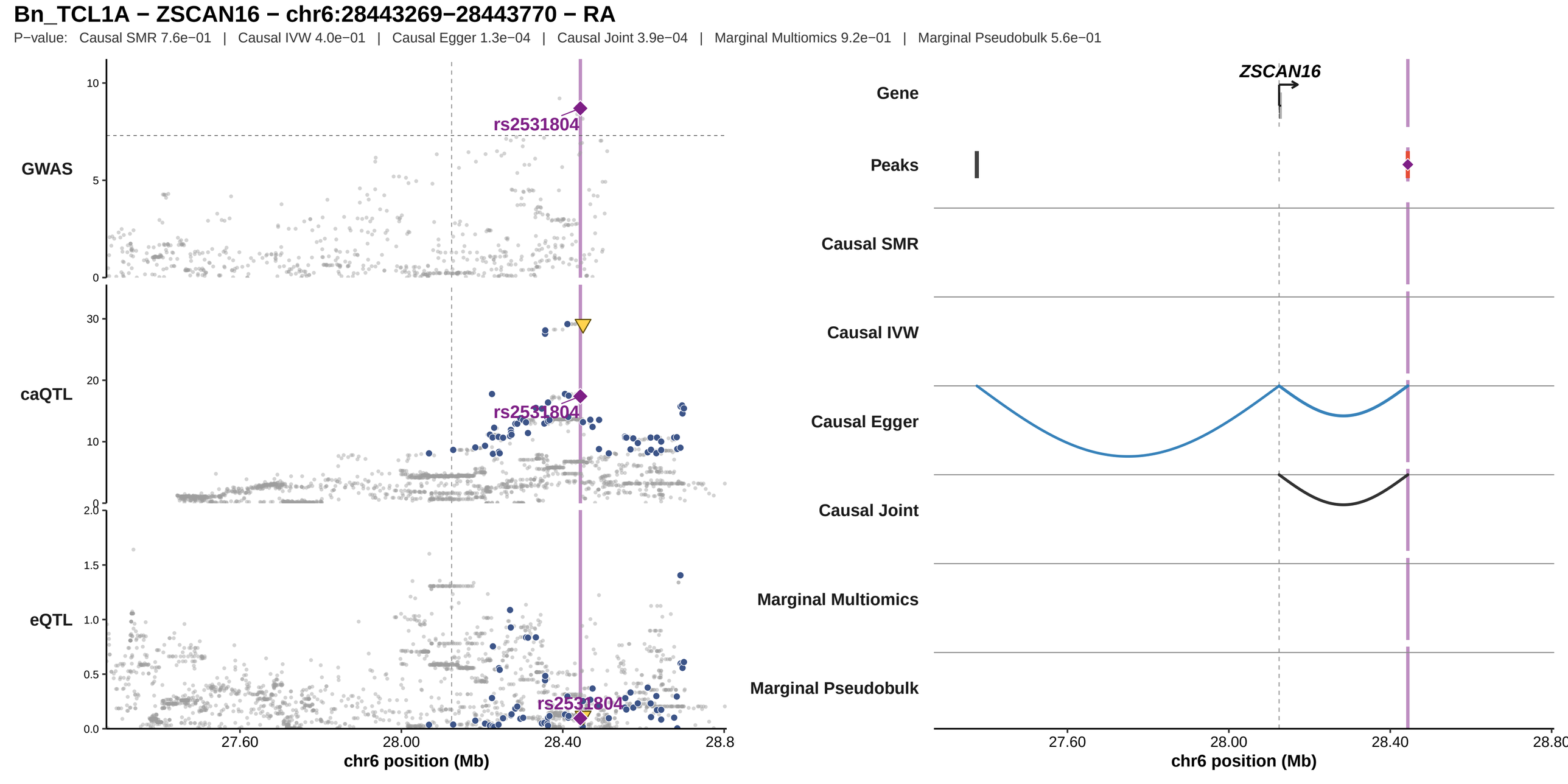
P−value: Causal SMR 2.9e−04 | Causal IVW 2.0e−04 | Causal Egger 9.0e−02 | Causal Joint 3.5e−04 | Marginal Multiomics 9.4e−01 | Marginal Pseudobulk 1.6e−01



Bn_TCL1A − ZKSCAN8 − chr6:28443269−28443770 − RA

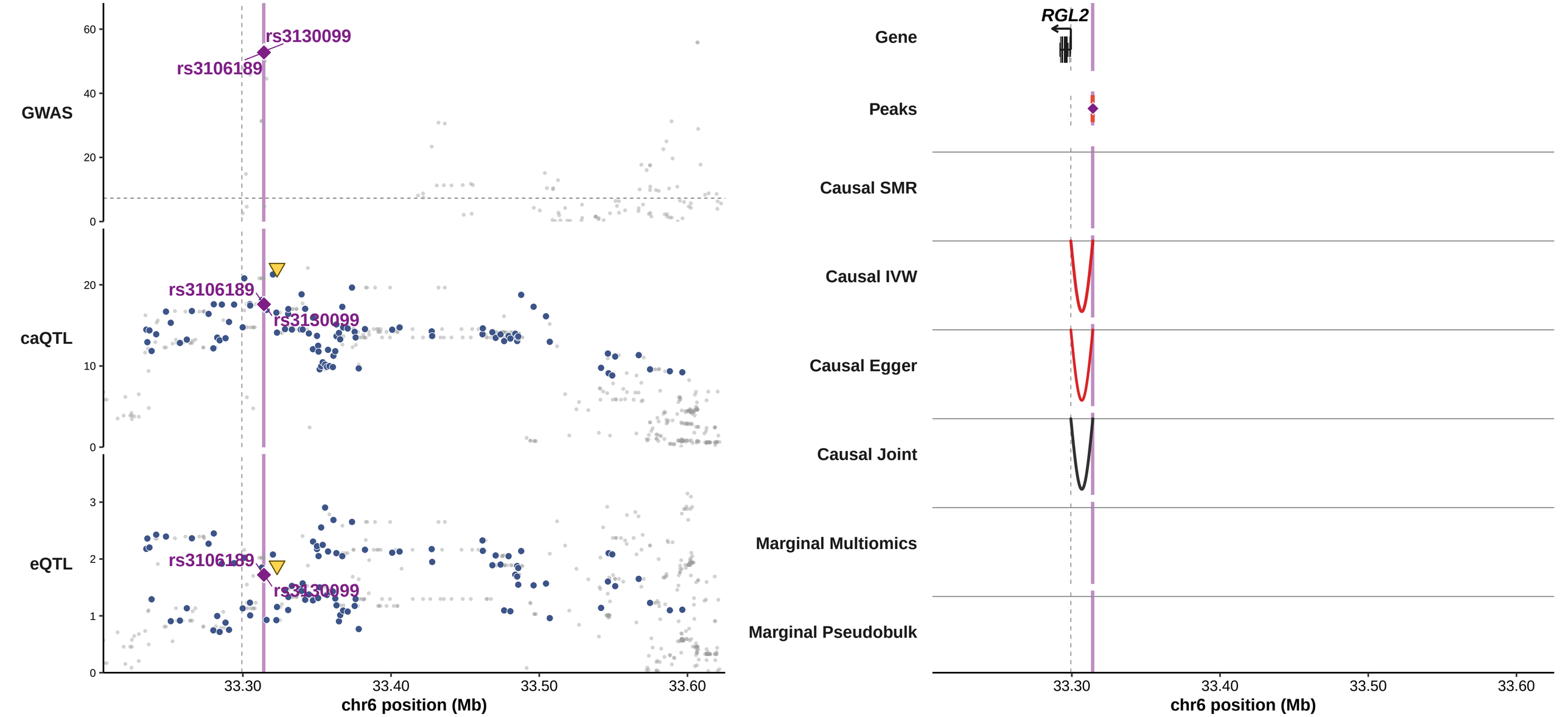
P−value: Causal SMR 1.8e−01 | Causal IVW 2.4e−03 | Causal Egger 1.1e−04 | Causal Joint 3.1e−04 | Marginal Multiomics 8.0e−01 | Marginal Pseudobulk 1.5e−01




Bn_TCL1A − ZSCAN16 − chr6:28443269−28443770 − RA
P−value: Causal SMR 7.6e−01 | Causal IVW 4.0e−01 | Causal Egger 1.3e−04 | Causal Joint 3.9e−04 | Marginal Multiomics 9.2e−01 | Marginal Pseudobulk 5.6e−01
GWAS
caQTL
eQTL
rs2531804
rs2531804
rs2531804
27.60
28.00
28.40
28.8
chr6 position (Mb)
Gene
ZSCAN16
Peaks
Causal SMR
Causal IVW
Causal Egger
Causal Joint
Marginal Multiomics
Marginal Pseudobulk
27.60
28.00
28.40
28.80
chr6 position (Mb)

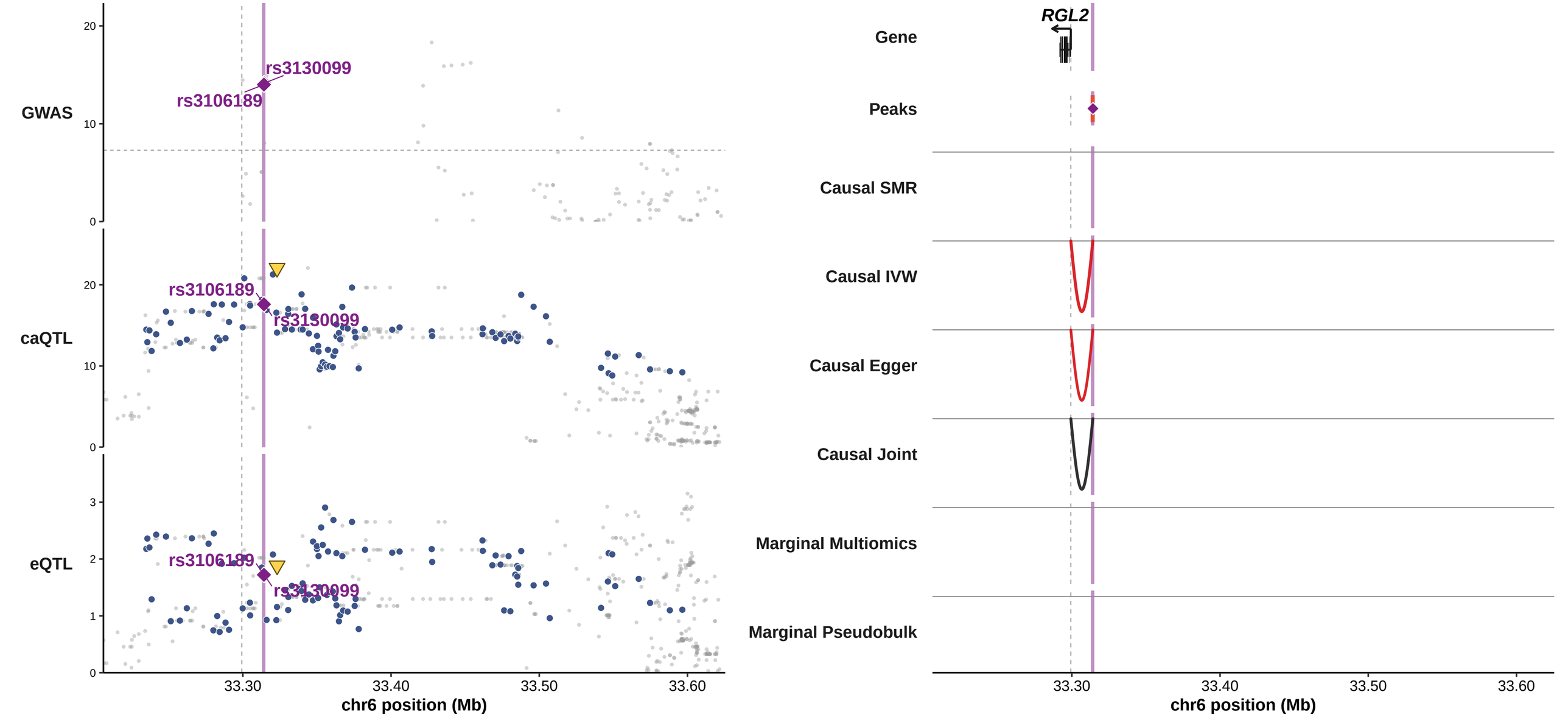
Bn_TCL1A − RGL2 − chr6:33313910−33314411 − T1D

P−value: Causal SMR 1.6e−02 | Causal IVW 9.1e−05 | Causal Egger 2.2e−03 | Causal Joint 2.6e−04 | Marginal Multiomics 2.6e−01 | Marginal Pseudobulk 5.9e−01



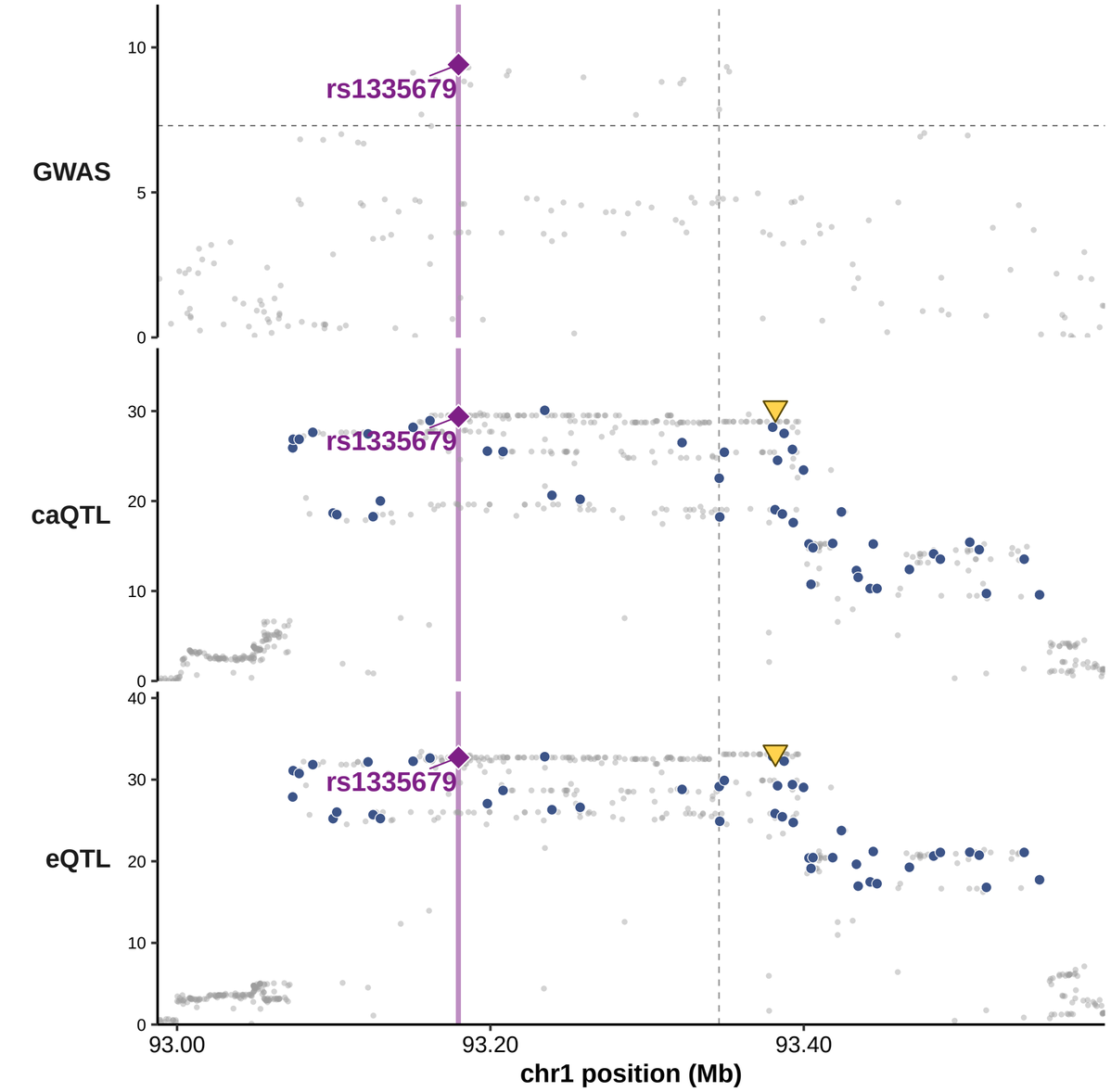
Bn_TCL1A − RGL2 − chr6:33313910−33314411 − RA

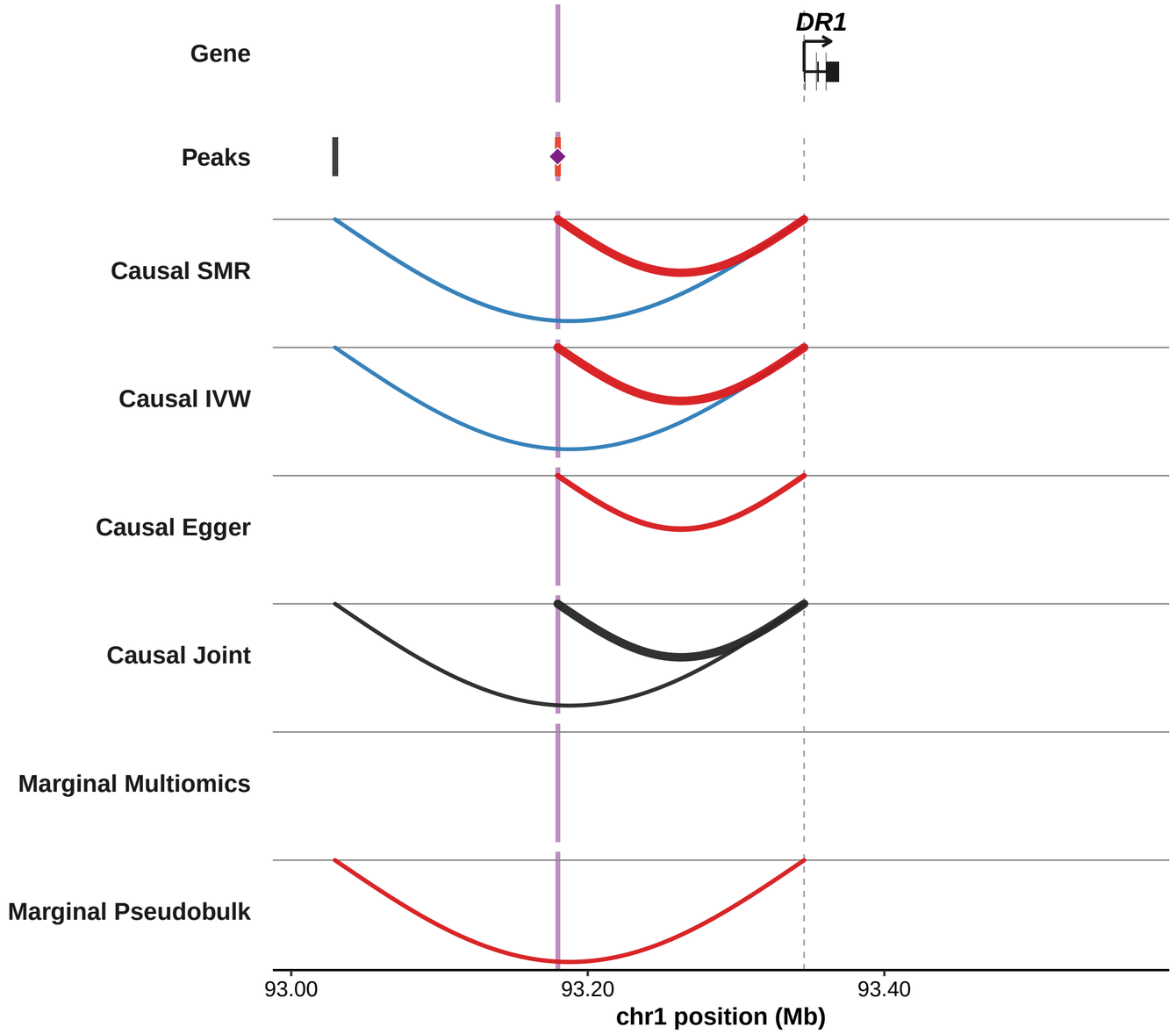
P−value: Causal SMR 1.6e−02 | Causal IVW 9.1e−05 | Causal Egger 2.2e−03 | Causal Joint 2.6e−04 | Marginal Multiomics 2.6e−01 | Marginal Pseudobulk 5.9e−01



CD4_Tn_CCR7 − DR1 − chr1:93179440−93179941 − CD

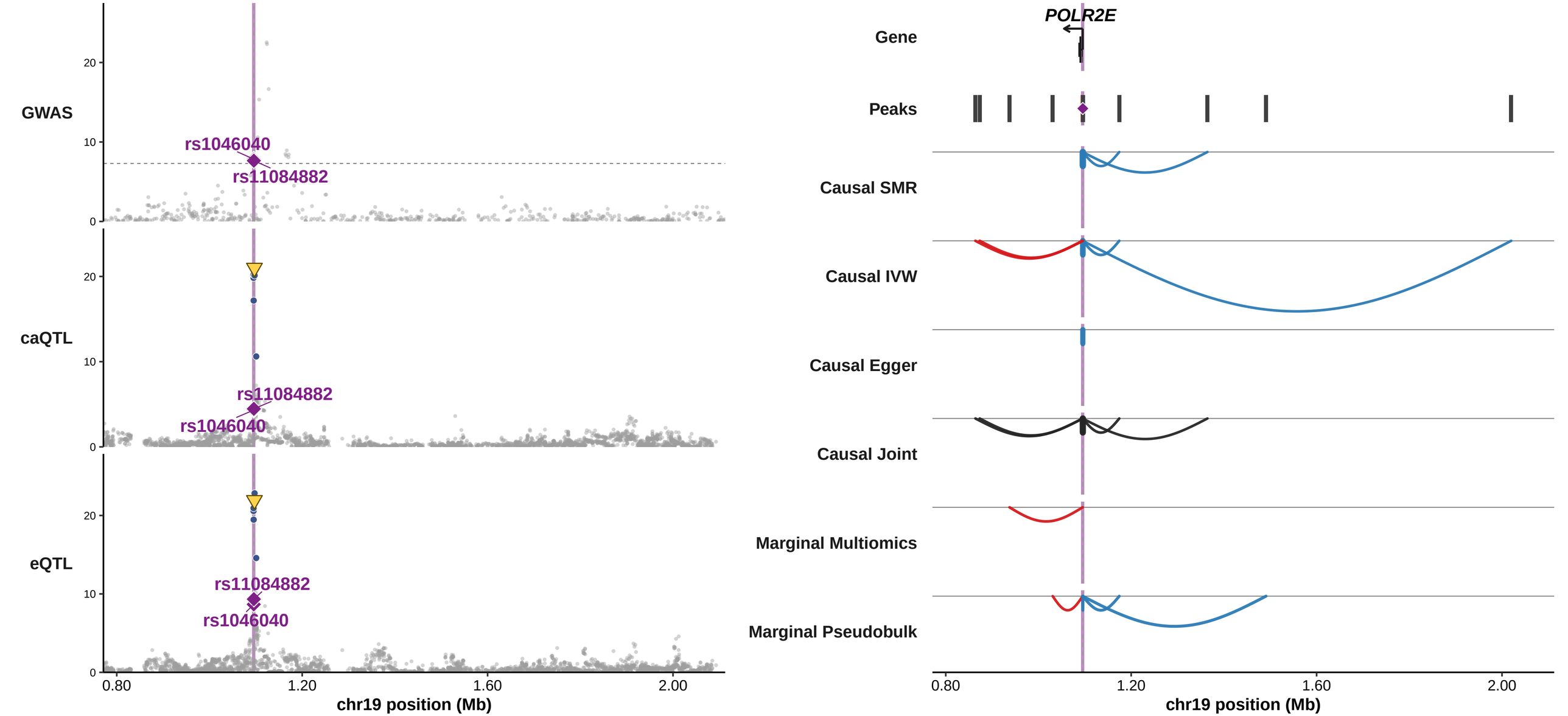
P−value: Causal SMR 4.8e−17 | Causal IVW 7.5e−18 | Causal Egger 1.1e−08 | Causal Joint 2.0e−17 | Marginal Multiomics 9.4e−01 | Marginal Pseudobulk 2.0e−01





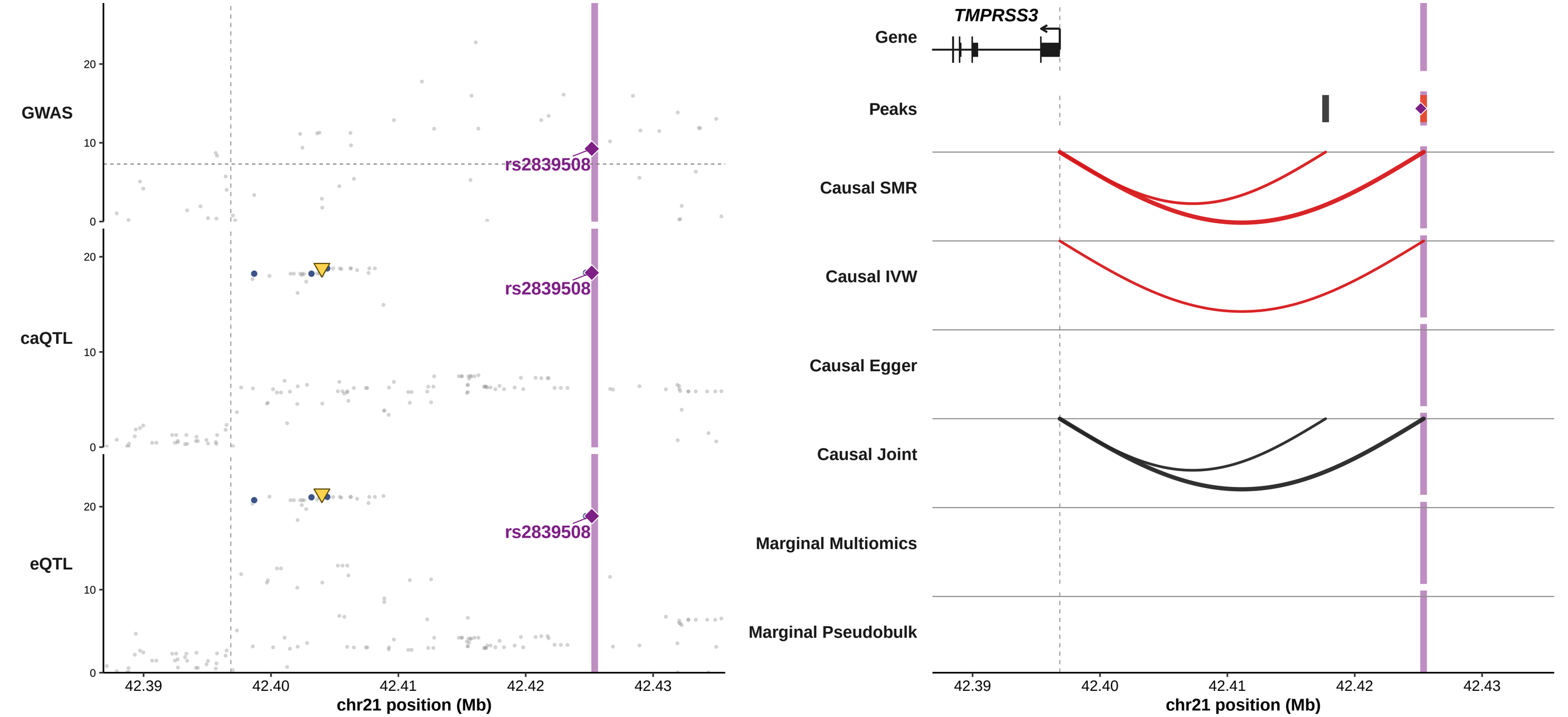
CD4_Tn_CCR7 − POLR2E − chr19:1095225−1095726 − CD

P−value: Causal SMR 7.3e−12 | Causal IVW 7.3e−05 | Causal Egger 2.0e−02 | Causal Joint 2.2e−11 | Marginal Multiomics 9.4e−01 | Marginal Pseudobulk 1.3e−02



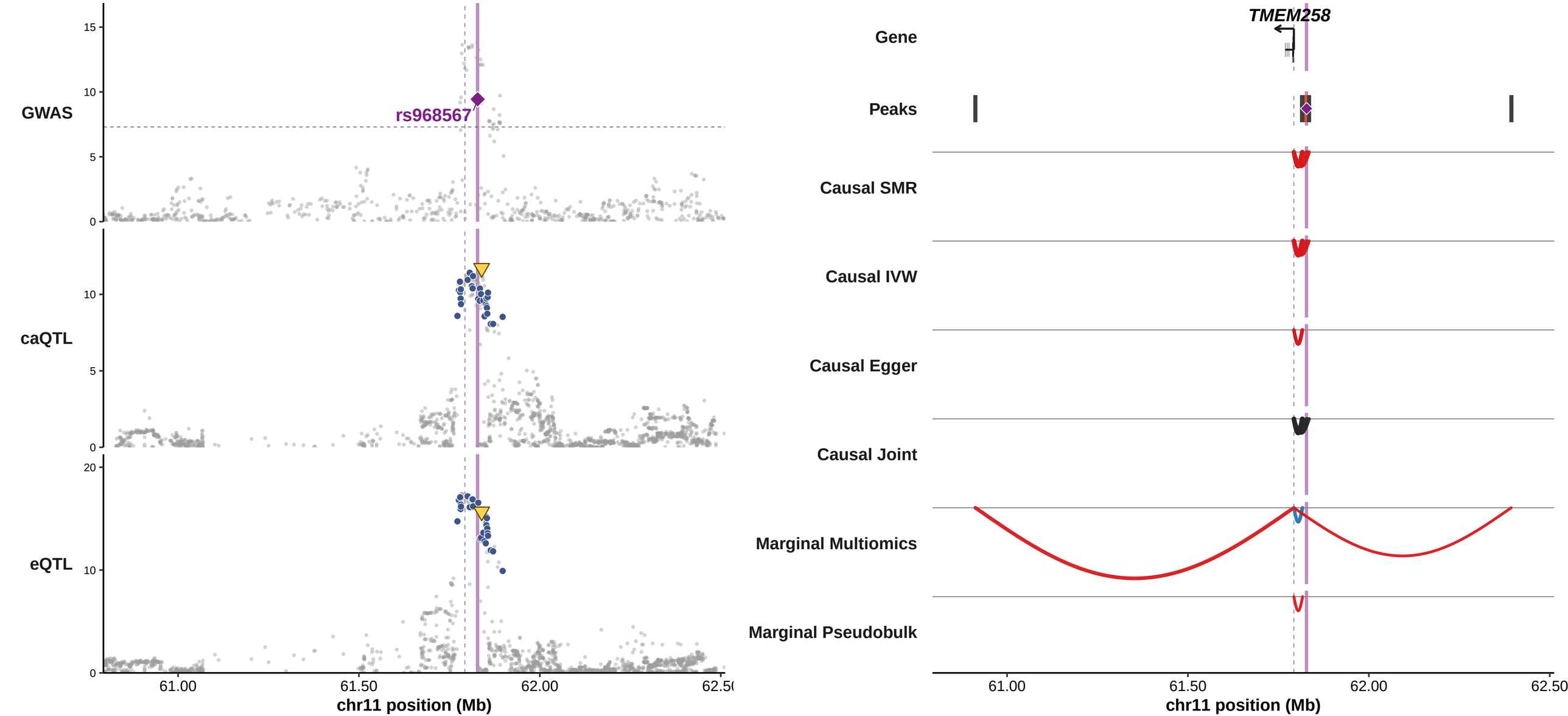
CD4_Tn_CCR7 − TMPRSS3 − chr21:42425157−42425658 − T1D

P−value: Causal SMR 3.6e−11 | Causal IVW 9.4e−04 | Causal Egger 2.0e−01 | Causal Joint 1.1e−10 | Marginal Multiomics 1.3e−01 | Marginal Pseudobulk 4.0e−02



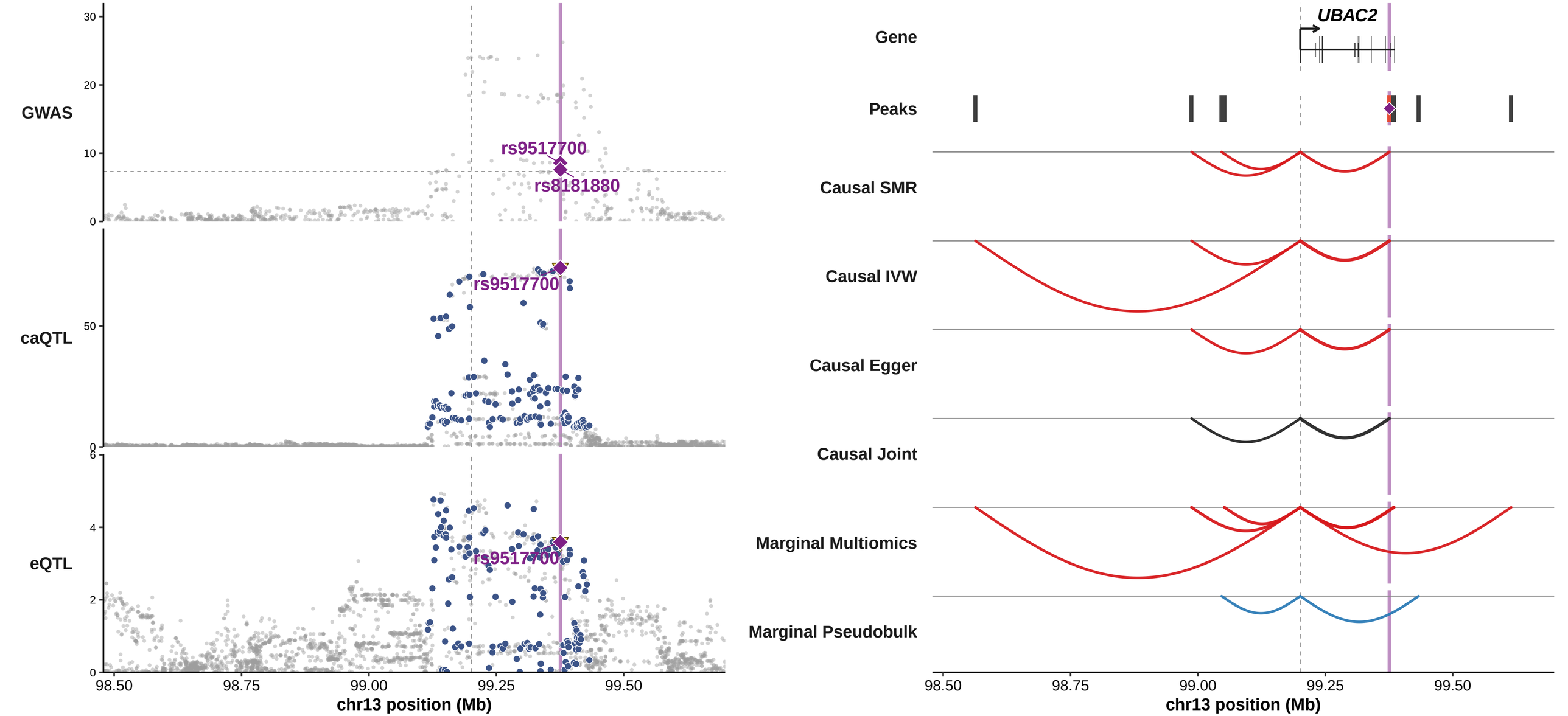
CD4_Tn_CCR7 − TMEM258 − chr11:61827740−61828241 − Asthma

P−value: Causal SMR 8.9e−08 | Causal IVW 3.1e−06 | Causal Egger 2.7e−01 | Causal Joint 2.6e−07 | Marginal Multiomics 3.0e−01 | Marginal Pseudobulk 8.4e−02



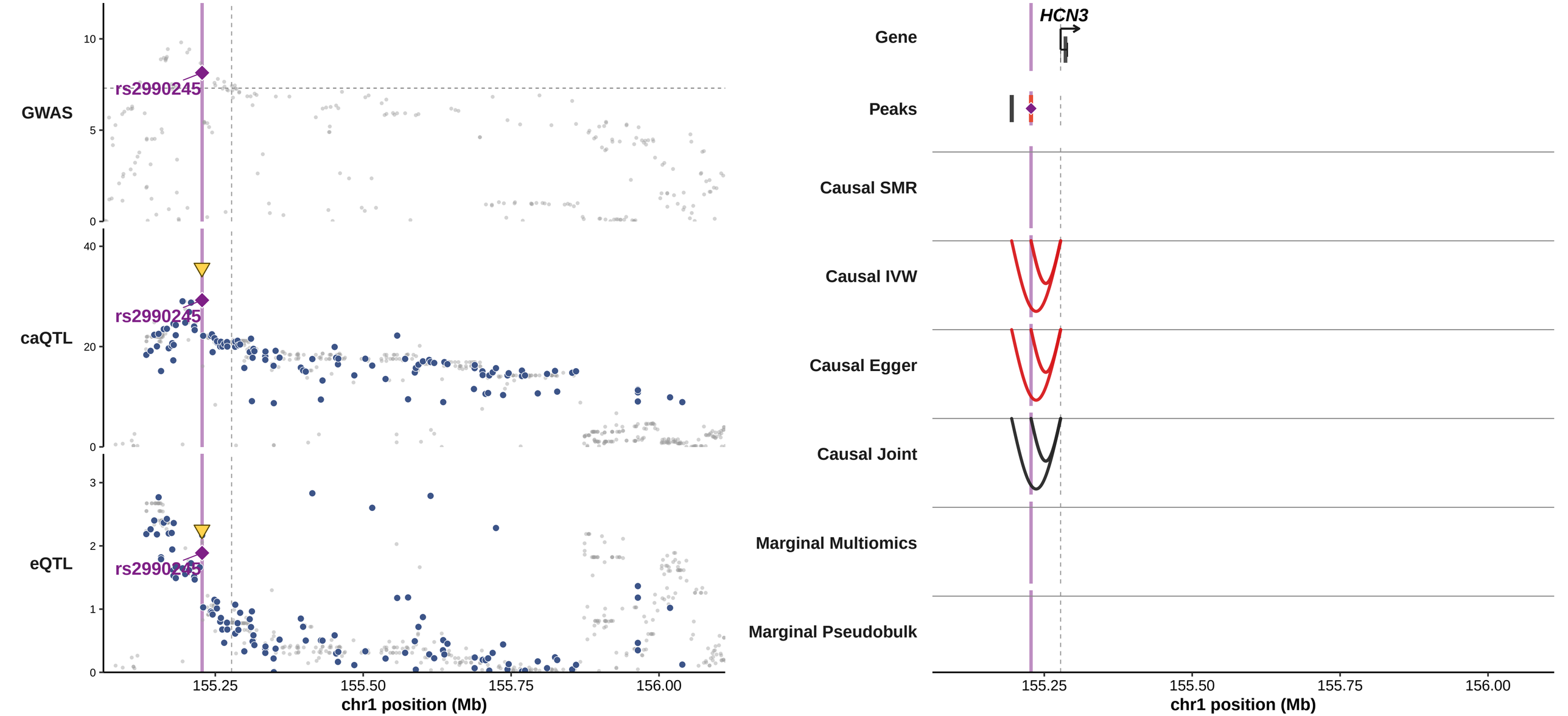
CD4_Tn_CCR7 − UBAC2 − chr13:99375239−99375740 − Asthma

P−value: Causal SMR 3.4e−04 | Causal IVW 7.8e−07 | Causal Egger 8.5e−06 | Causal Joint 2.1e−06 | Marginal Multiomics 4.5e−02 | Marginal Pseudobulk 3.7e−01



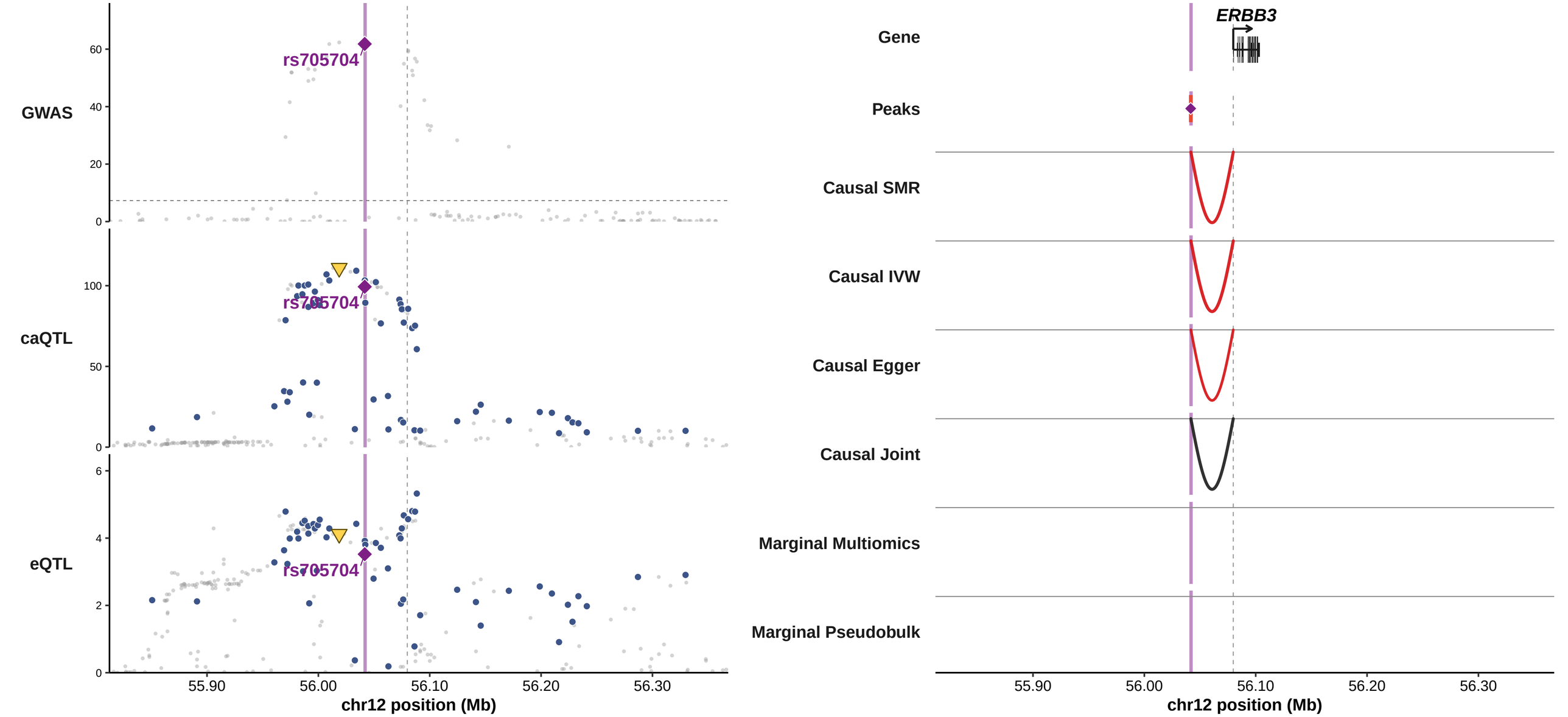
CD4_Tn_CCR7 − HCN3 − chr1:155227359−155227860 − IBD

P−value: Causal SMR 6.8e−03 | Causal IVW 1.4e−06 | Causal Egger 2.6e−06 | Causal Joint 2.8e−06 | Marginal Multiomics 1.9e−01 | Marginal Pseudobulk 1.0e+00



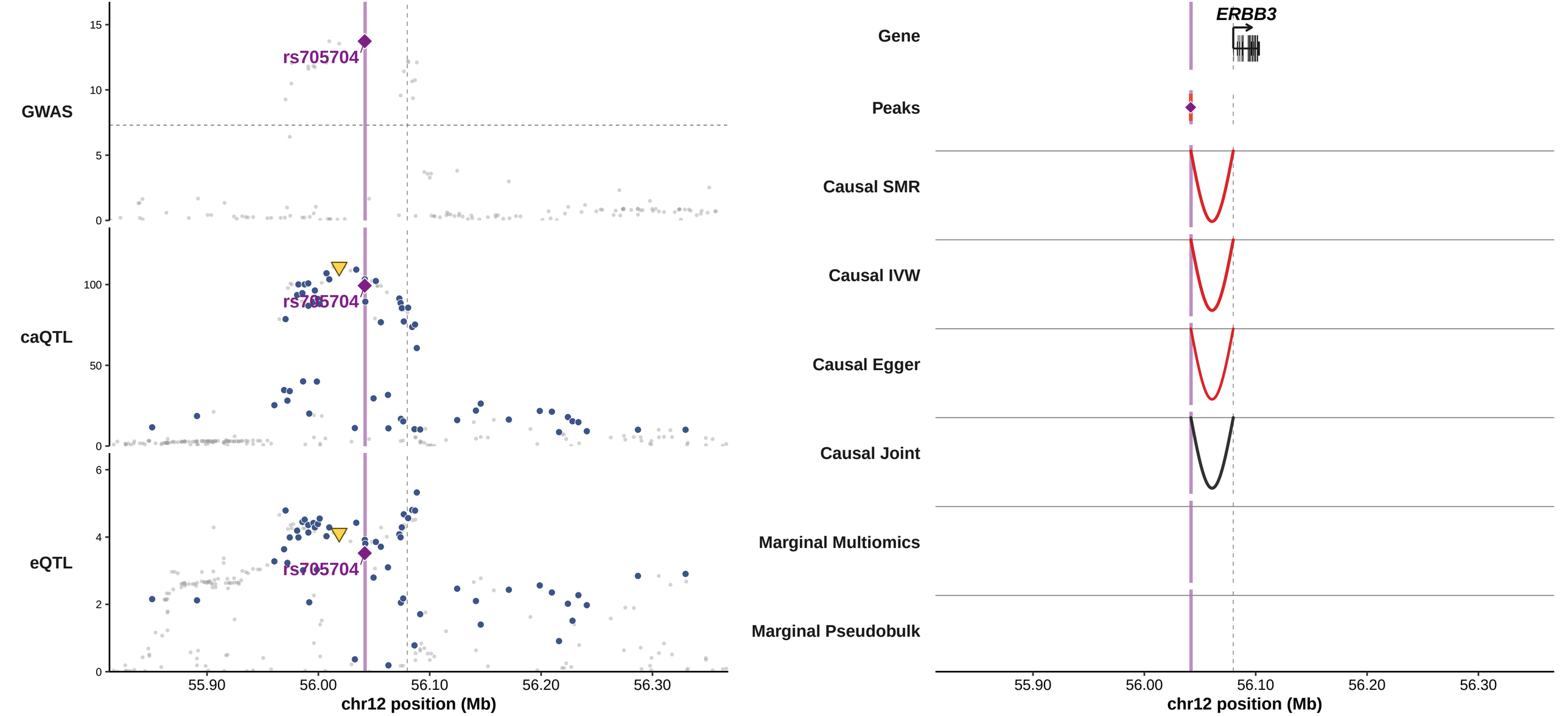
CD4_Tn_CCR7 − ERBB3 − chr12:56041553−56042054 − T1D

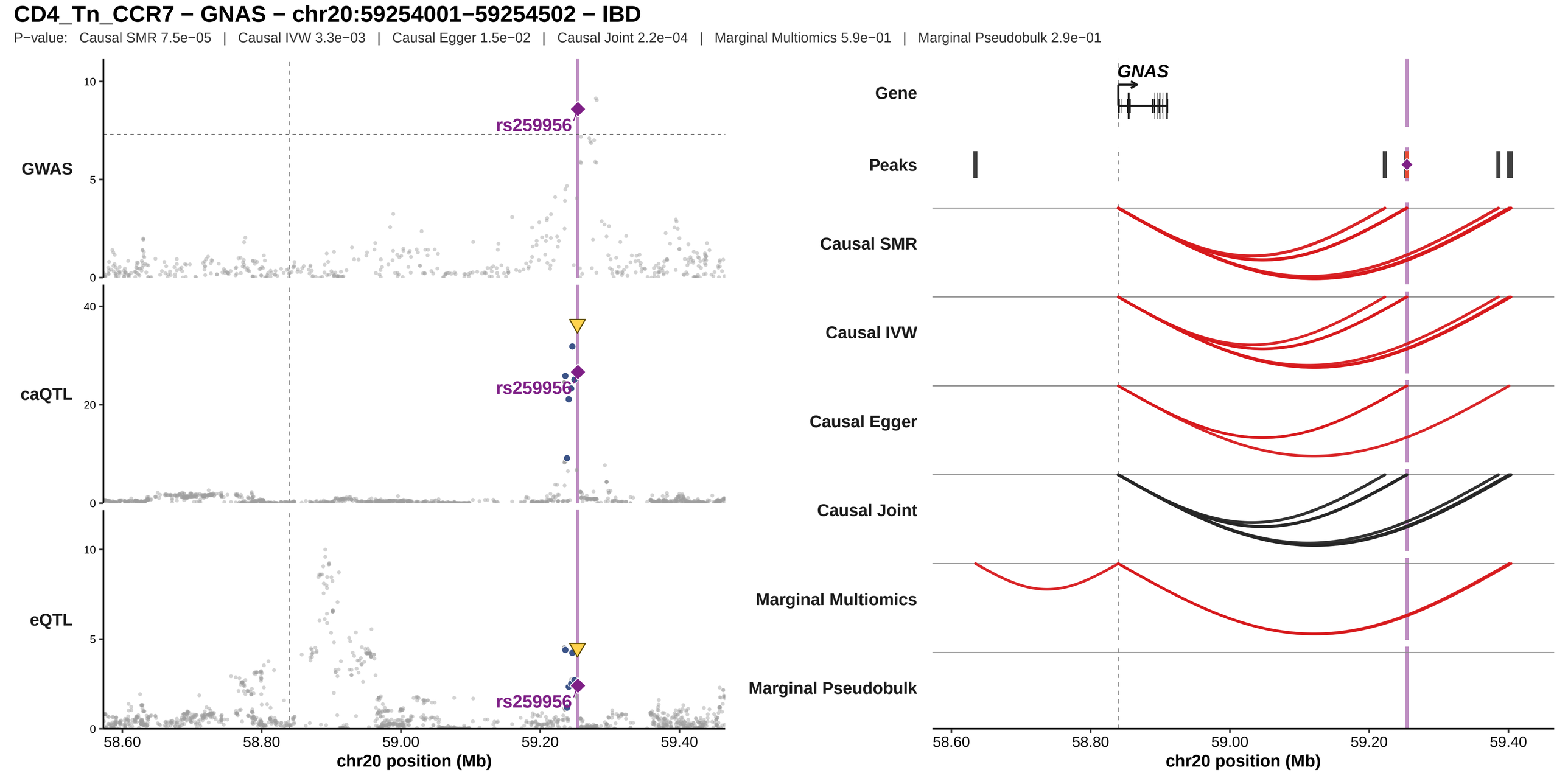
P−value: Causal SMR 9.7e−05 | Causal IVW 4.7e−05 | Causal Egger 4.4e−03 | Causal Joint 9.4e−05 | Marginal Multiomics 1.5e−01 | Marginal Pseudobulk 4.0e−02

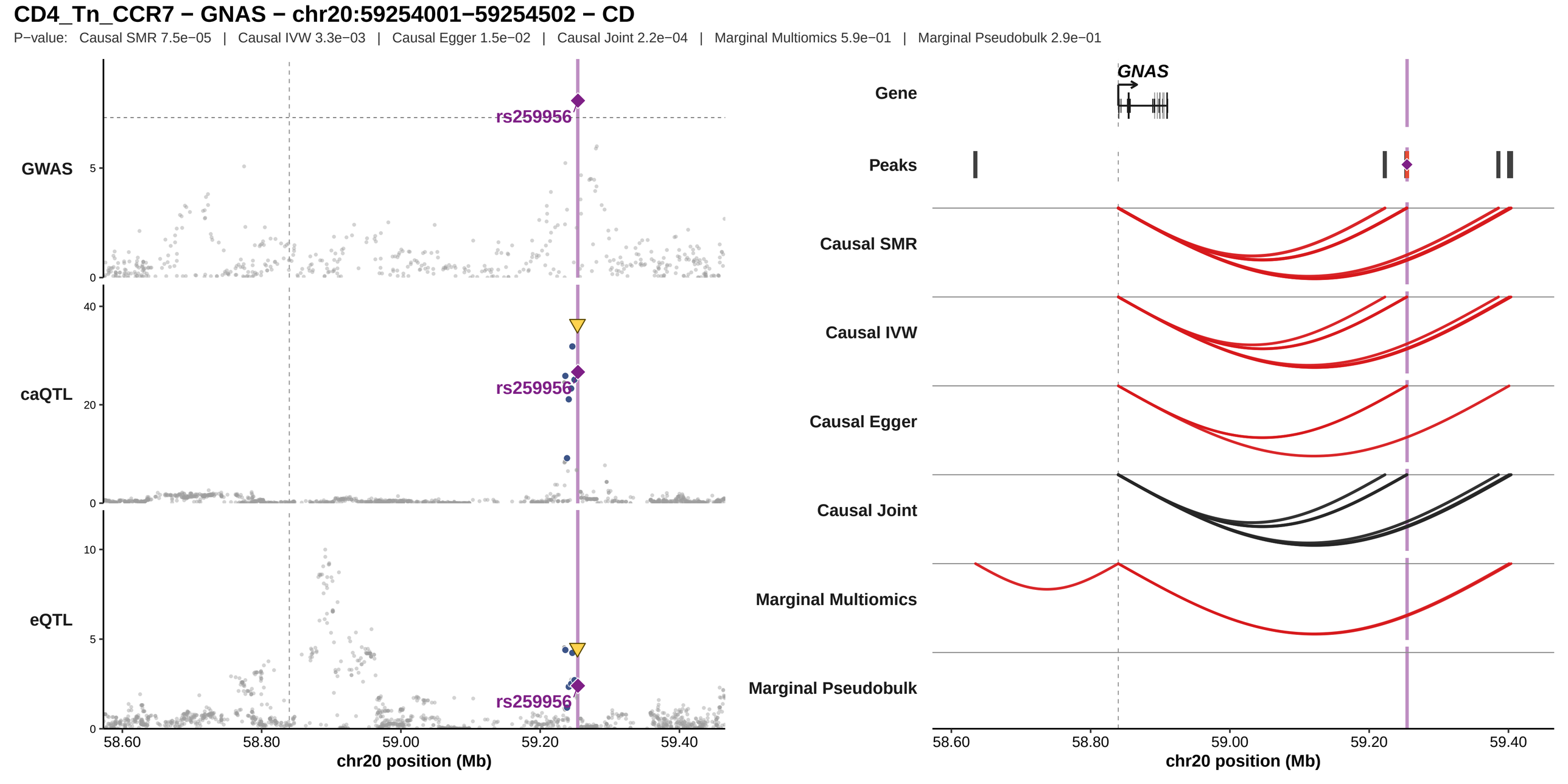


CD4_Tn_CCR7 − ERBB3 − chr12:56041553−56042054 − Asthma

P−value: Causal SMR 9.7e−05 | Causal IVW 4.7e−05 | Causal Egger 4.4e−03 | Causal Joint 9.4e−05 | Marginal Multiomics 1.5e−01 | Marginal Pseudobulk 4.0e−02

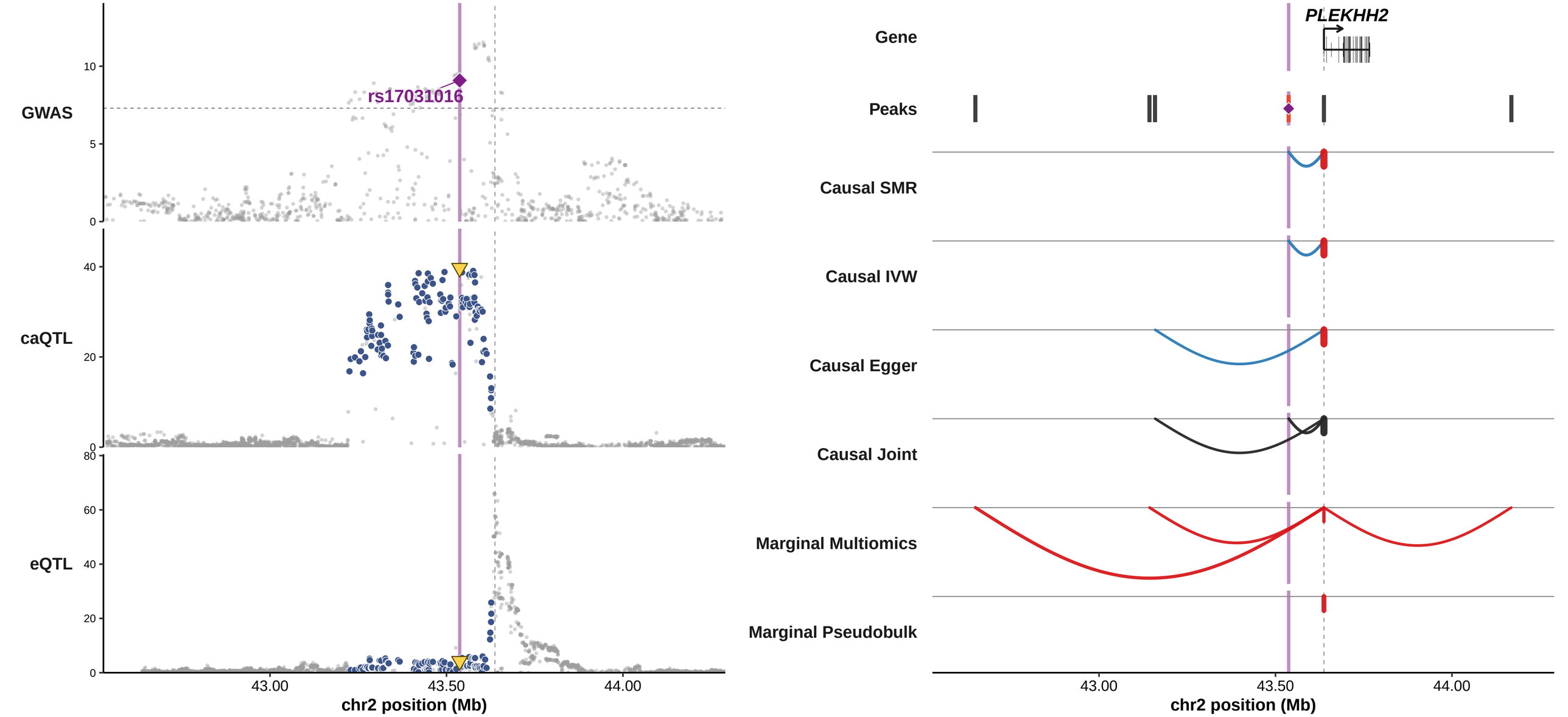



CD4_Tn_CCR7 − GNAS − chr20:59254001−59254502 − IBD
P−value: Causal SMR 7.5e−05 | Causal IVW 3.3e−03 | Causal Egger 1.5e−02 | Causal Joint 2.2e−04 | Marginal Multiomics 5.9e−01 | Marginal Pseudobulk 2.9e−01
GWAS
caQTL
eQTL
rs259956
rs259956
rs259956
10
5
0
40
20
0
10
5
0
58.60
58.80
59.00
59.20
59.40
chr20 position (Mb)
Gene
GNAS
Peaks
Causal SMR
Causal IVW
Causal Egger
Causal Joint
Marginal Multiomics
Marginal Pseudobulk
58.60
58.80
59.00
59.20
59.40
chr20 position (Mb)


CD4_Tn_CCR7 − GNAS − chr20:59254001−59254502 − CD
P−value: Causal SMR 7.5e−05 | Causal IVW 3.3e−03 | Causal Egger 1.5e−02 | Causal Joint 2.2e−04 | Marginal Multiomics 5.9e−01 | Marginal Pseudobulk 2.9e−01
GWAS
caQTL
eQTL
rs259956
rs259956
rs259956
0
5
0
40
20
0
10
5
0
58.60
58.80
59.00
59.20
59.40
chr20 position (Mb)
Gene
GNAS
Peaks
Causal SMR
Causal IVW
Causal Egger
Causal Joint
Marginal Multiomics
Marginal Pseudobulk
58.60
58.80
59.00
59.20
59.40
chr20 position (Mb)

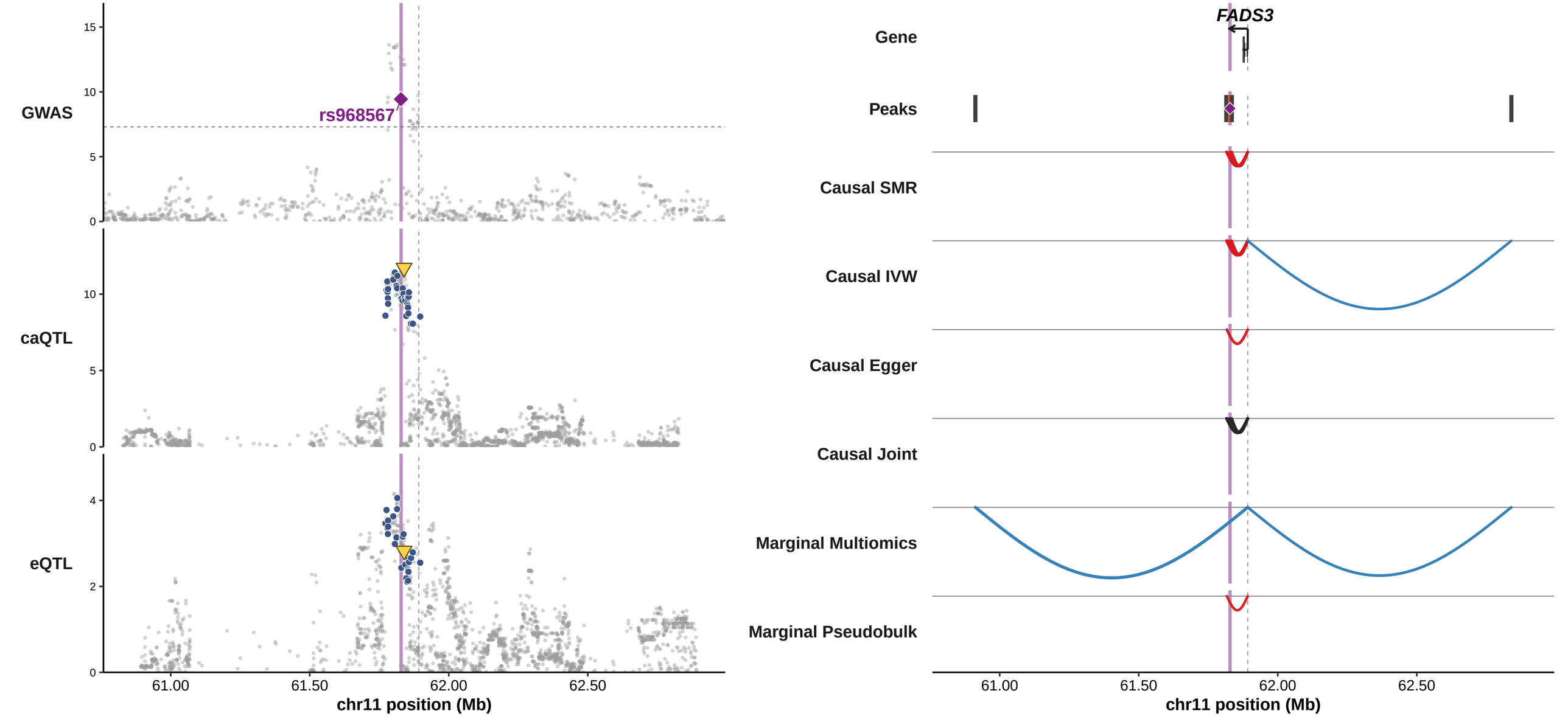
CD4_Tn_CCR7 − PLEKHH2 − chr2:43537114−43537615 − CD

P−value: Causal SMR 1.1e−04 | Causal IVW 2.7e−04 | Causal Egger 2.7e−01 | Causal Joint 2.3e−04 | Marginal Multiomics 1.1e−01 | Marginal Pseudobulk 4.1e−02



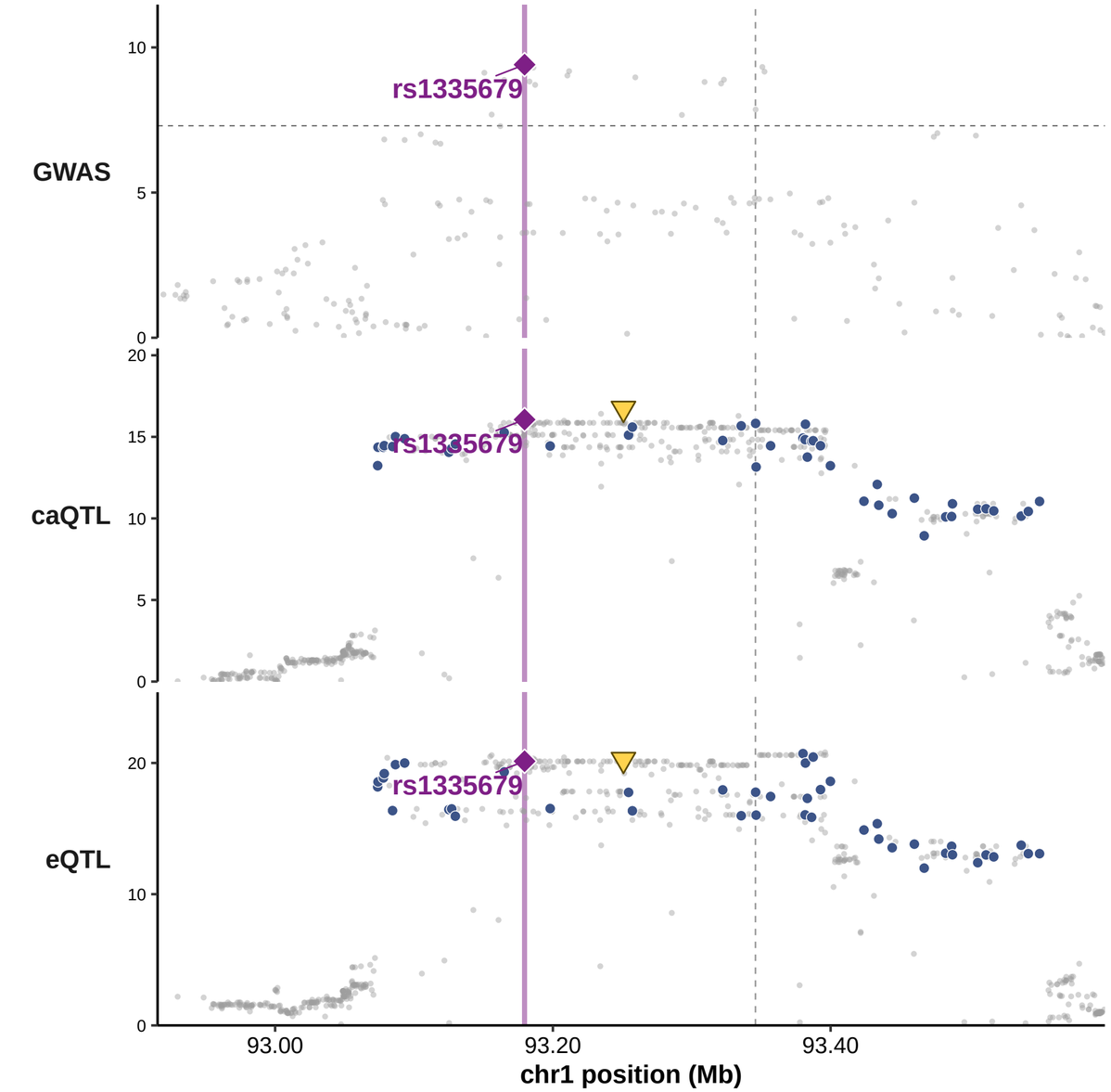
CD4_Tn_CCR7 − FADS3 − chr11:61827740−61828241 − Asthma

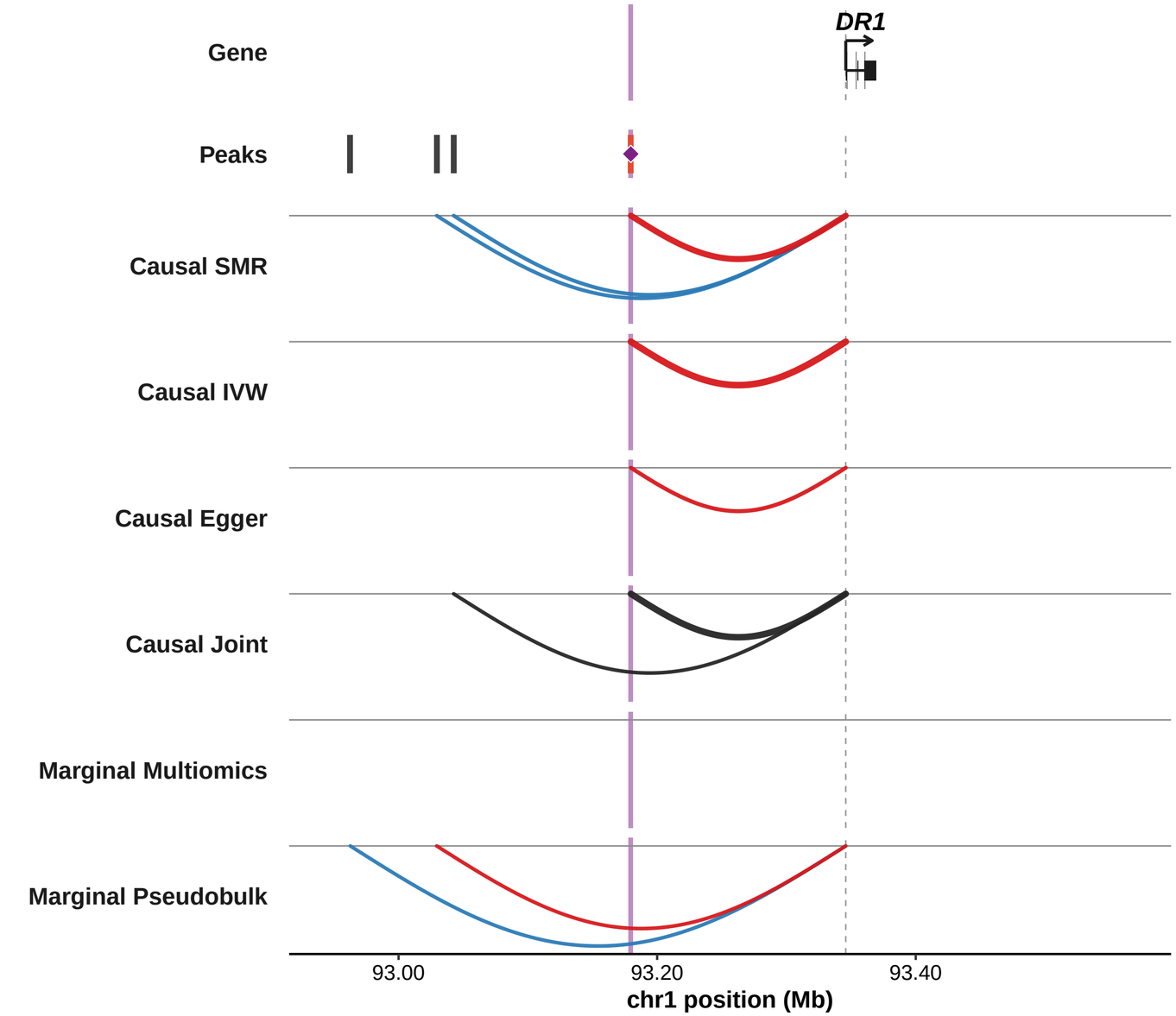
P−value: Causal SMR 3.8e−03 | Causal IVW 1.3e−04 | Causal Egger 7.1e−02 | Causal Joint 3.9e−04 | Marginal Multiomics 9.8e−01 | Marginal Pseudobulk 2.8e−01



CD8_Tn_CCR7 − DR1 − chr1:93179440−93179941 − CD

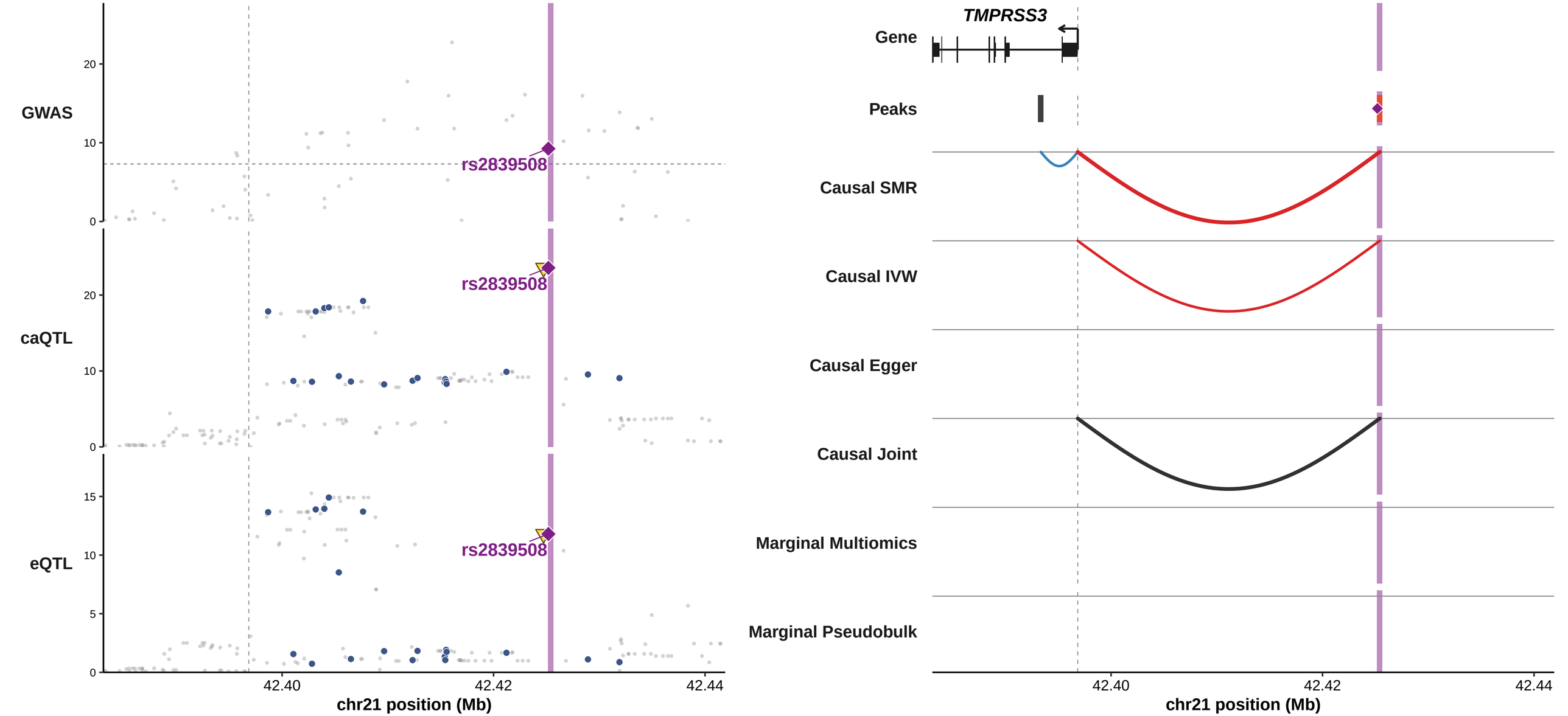
P−value: Causal SMR 2.9e−10 | Causal IVW 3.8e−12 | Causal Egger 5.9e−04 | Causal Joint 1.1e−11 | Marginal Multiomics 9.4e−01 | Marginal Pseudobulk 6.5e−01





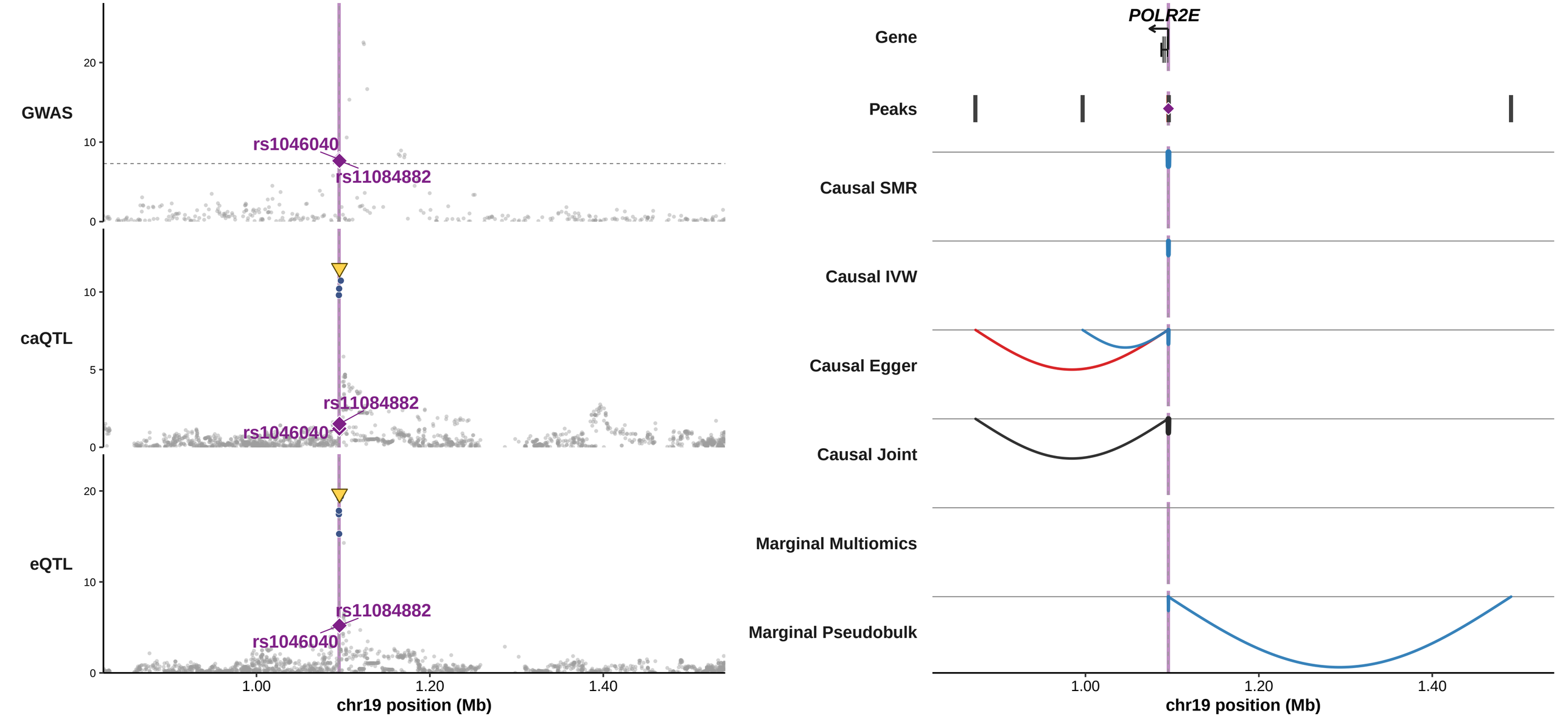
CD8_Tn_CCR7 − TMPRSS3 − chr21:42425157−42425658 − T1D

P−value: Causal SMR 6.5e−09 | Causal IVW 2.1e−03 | Causal Egger 2.3e−01 | Causal Joint 2.0e−08 | Marginal Multiomics 1.3e−01 | Marginal Pseudobulk 5.3e−02



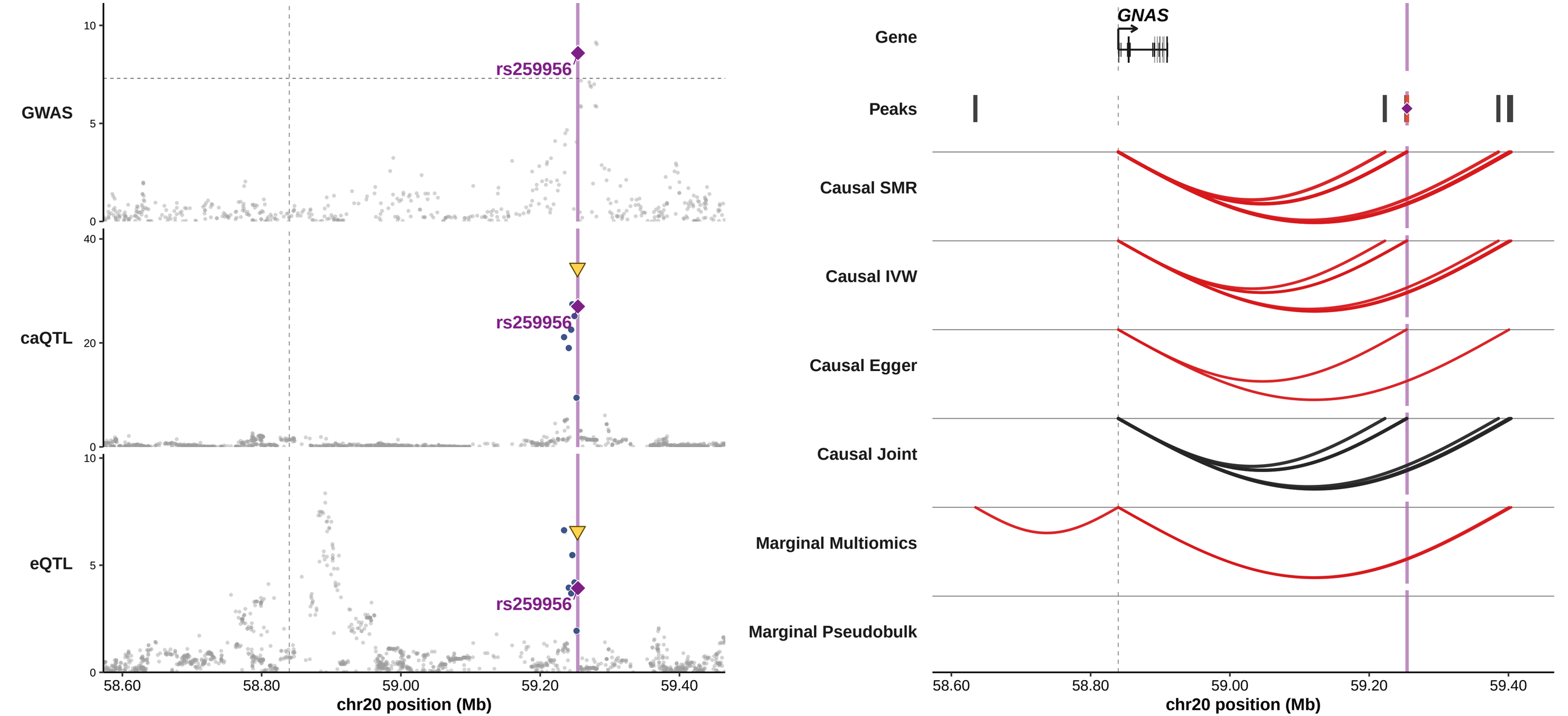
CD8_Tn_CCR7 − POLR2E − chr19:1095225−1095726 − CD

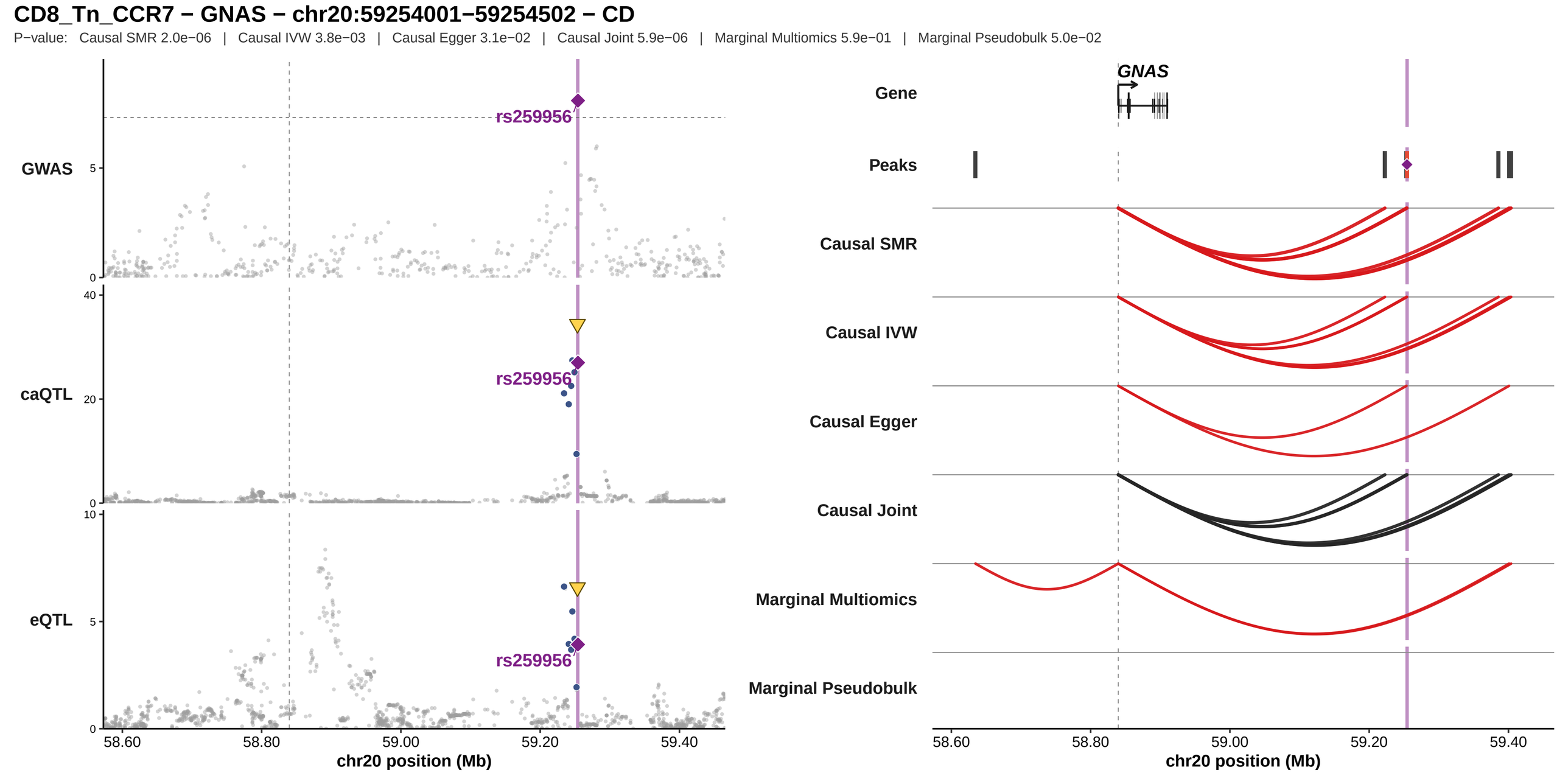
P−value: Causal SMR 2.5e−08 | Causal IVW 7.9e−04 | Causal Egger 8.4e−02 | Causal Joint 7.4e−08 | Marginal Multiomics 9.4e−01 | Marginal Pseudobulk 9.8e−01



CD8_Tn_CCR7 − GNAS − chr20:59254001−59254502 − IBD

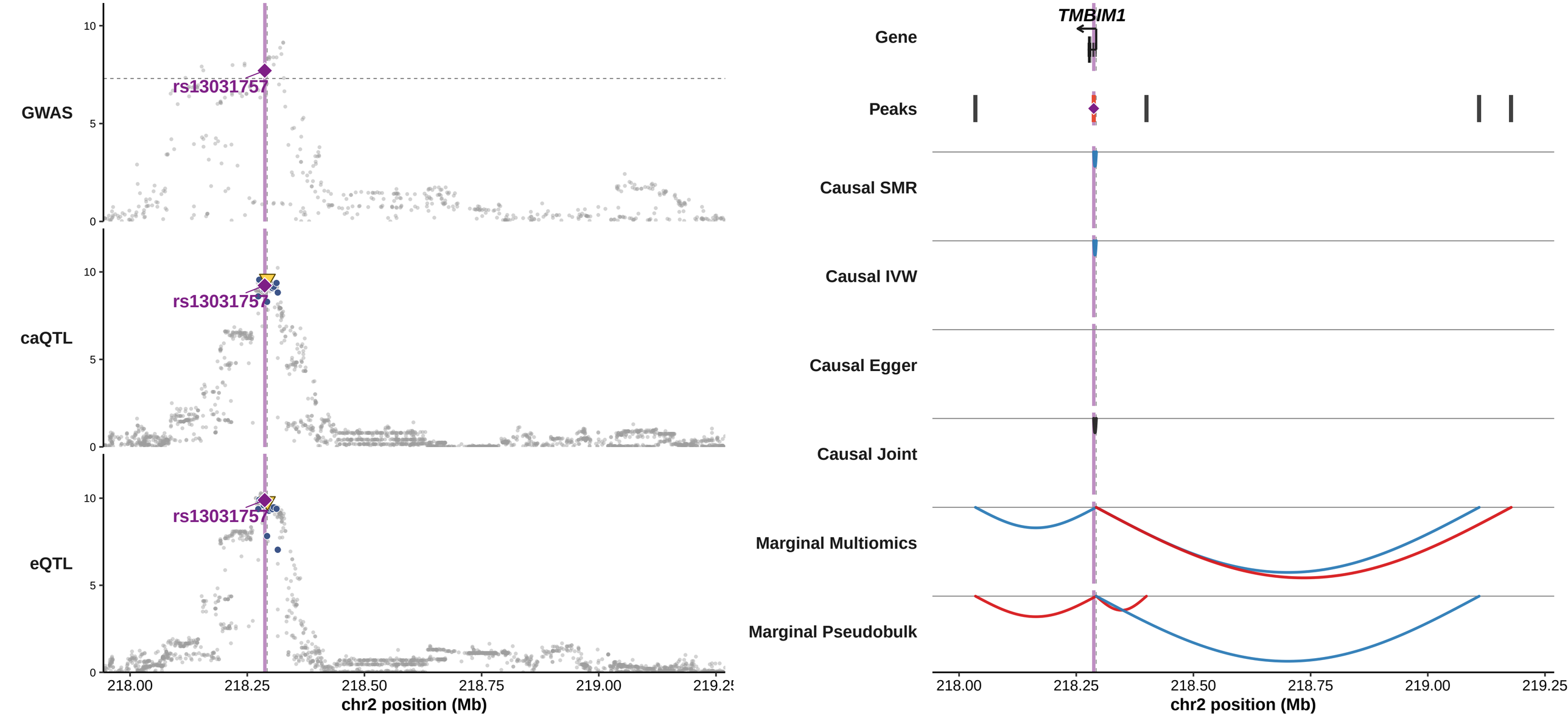
P−value: Causal SMR 2.0e−06 | Causal IVW 3.8e−03 | Causal Egger 3.1e−02 | Causal Joint 5.9e−06 | Marginal Multiomics 5.9e−01 | Marginal Pseudobulk 5.0e−02




CD8_Tn_CCR7 − GNAS − chr20:59254001−59254502 − CD
P−value: Causal SMR 2.0e−06 | Causal IVW 3.8e−03 | Causal Egger 3.1e−02 | Causal Joint 5.9e−06 | Marginal Multiomics 5.9e−01 | Marginal Pseudobulk 5.0e−02
GWAS
caQTL
eQTL
rs259956
rs259956
rs259956
0
5
40
20
10
5
0
58.60
58.80
59.00
59.20
59.40
chr20 position (Mb)
Gene
GNAS
Peaks
Causal SMR
Causal IVW
Causal Egger
Causal Joint
Marginal Multiomics
Marginal Pseudobulk
58.60
58.80
59.00
59.20
59.40
chr20 position (Mb)

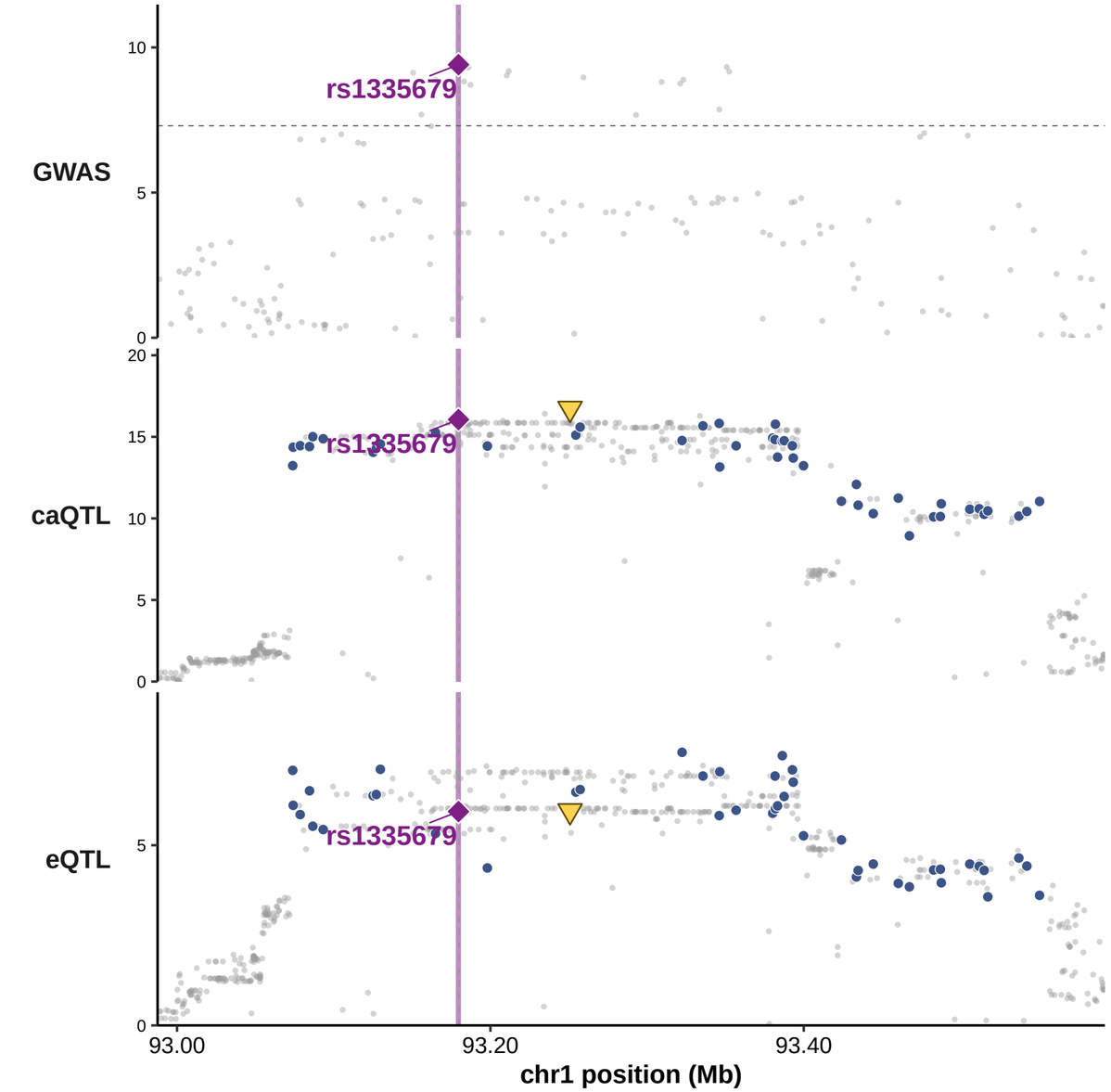
CD8_Tn_CCR7 − TMBIM1 − chr2:218287070−218287571 − CD

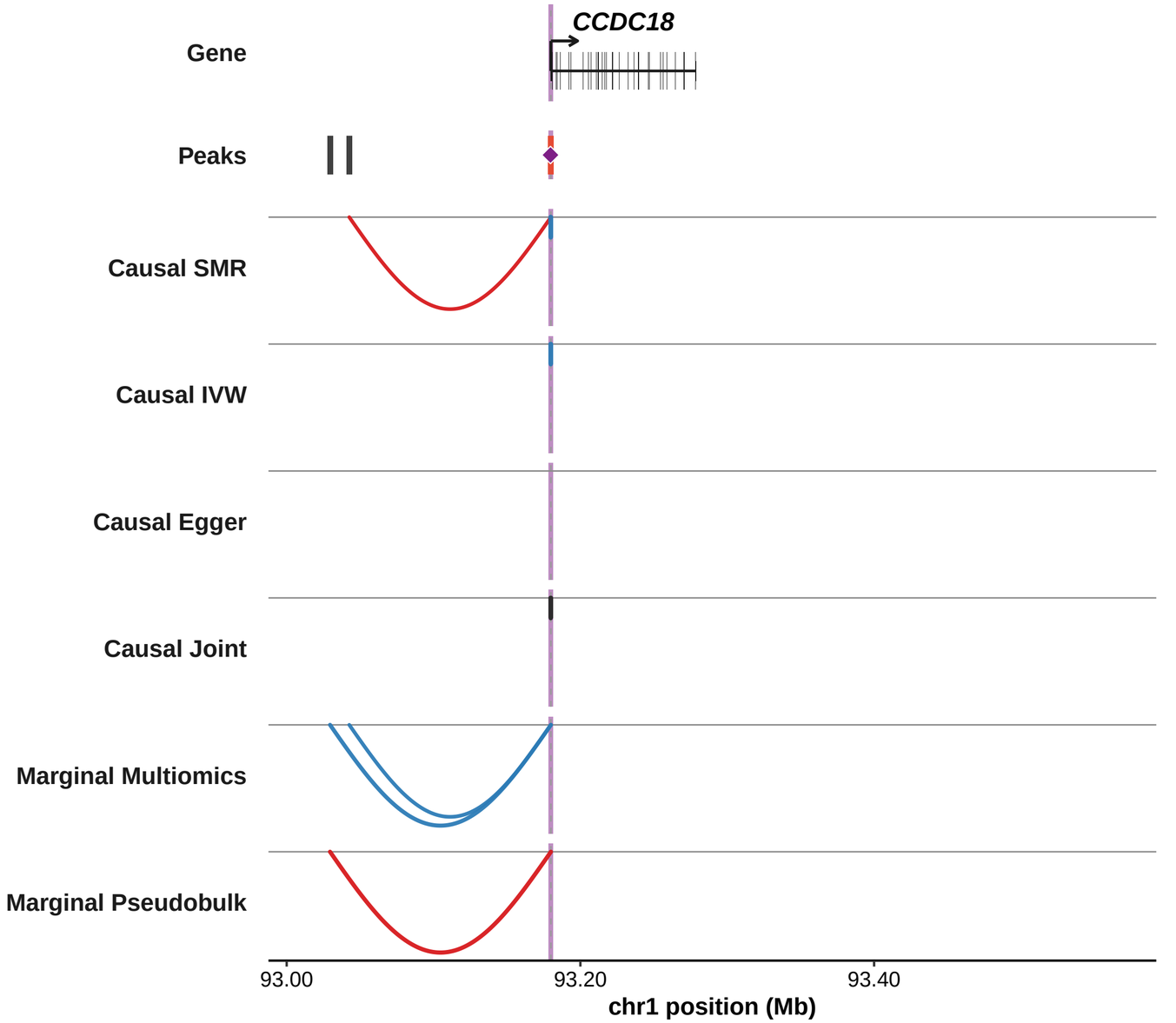
P−value: Causal SMR 6.7e−06 | Causal IVW 3.2e−05 | Causal Egger 8.8e−02 | Causal Joint 1.7e−05 | Marginal Multiomics 1.7e−01 | Marginal Pseudobulk 7.8e−02



CD8_Tn_CCR7 − CCDC18 − chr1:93179440−93179941 − CD

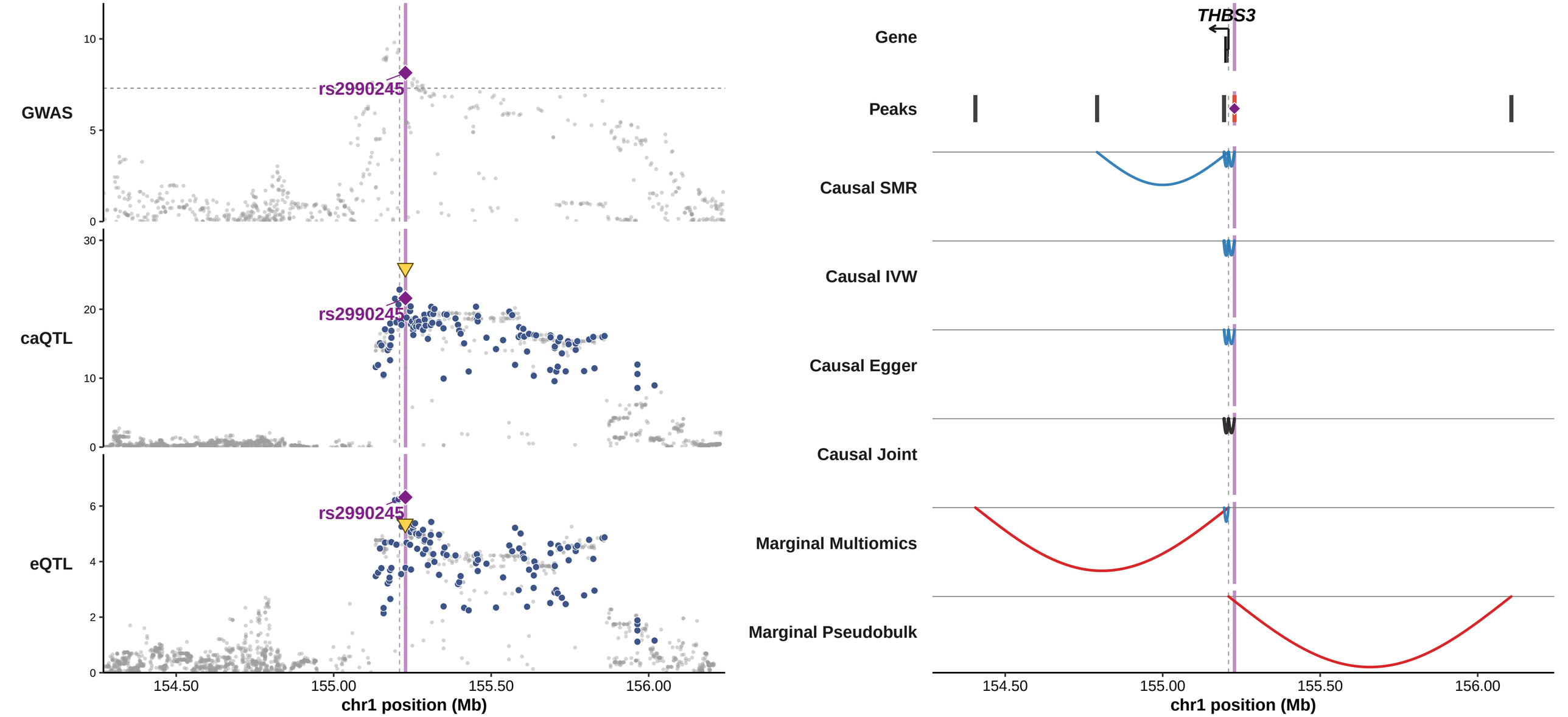
P−value: Causal SMR 2.4e−05 | Causal IVW 5.2e−05 | Causal Egger 2.1e−01 | Causal Joint 4.9e−05 | Marginal Multiomics 3.4e−01 | Marginal Pseudobulk 1.6e−01





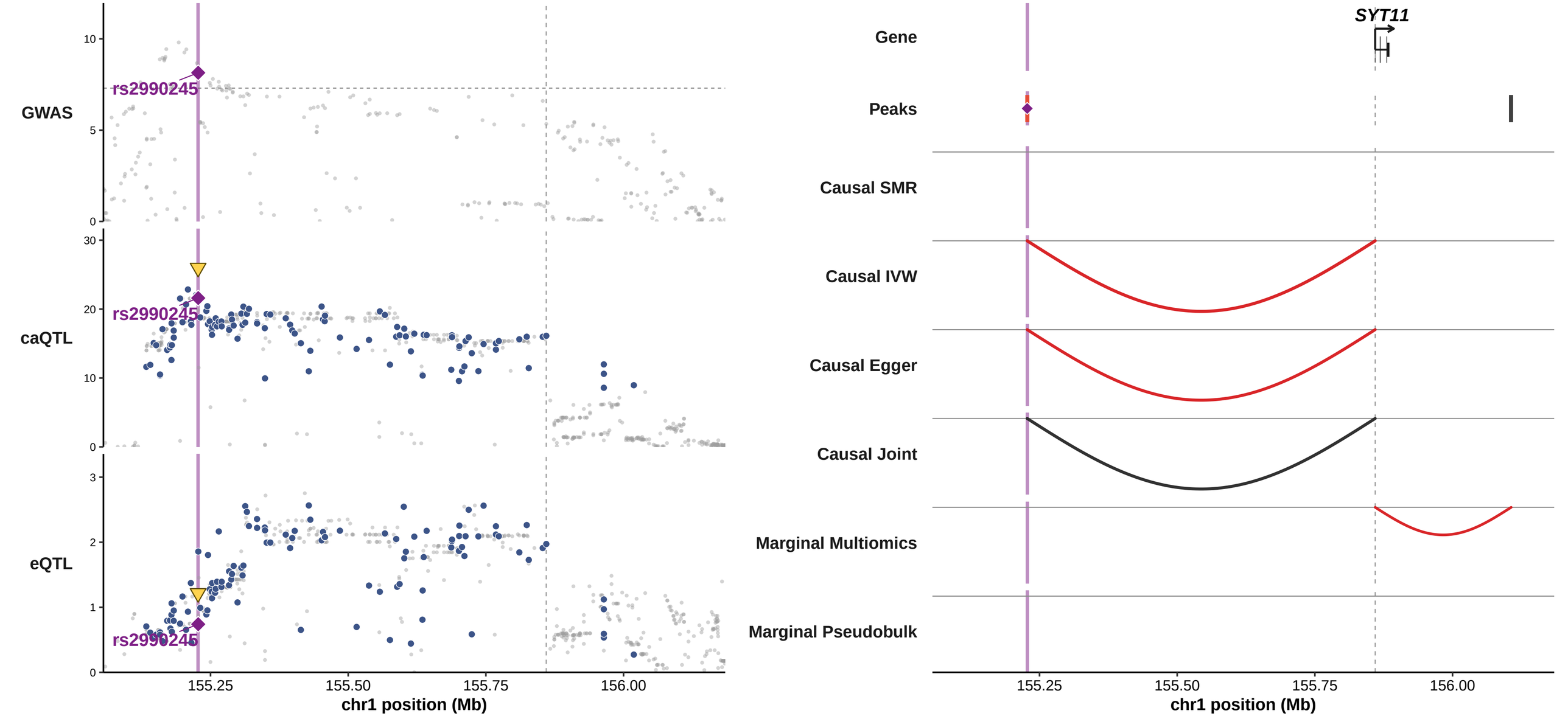
CD8_Tn_CCR7 − THBS3 − chr1:155227359−155227860 − IBD

P−value: Causal SMR 2.4e−05 | Causal IVW 5.1e−05 | Causal Egger 4.2e−03 | Causal Joint 4.9e−05 | Marginal Multiomics 1.8e−02 | Marginal Pseudobulk 1.7e−01



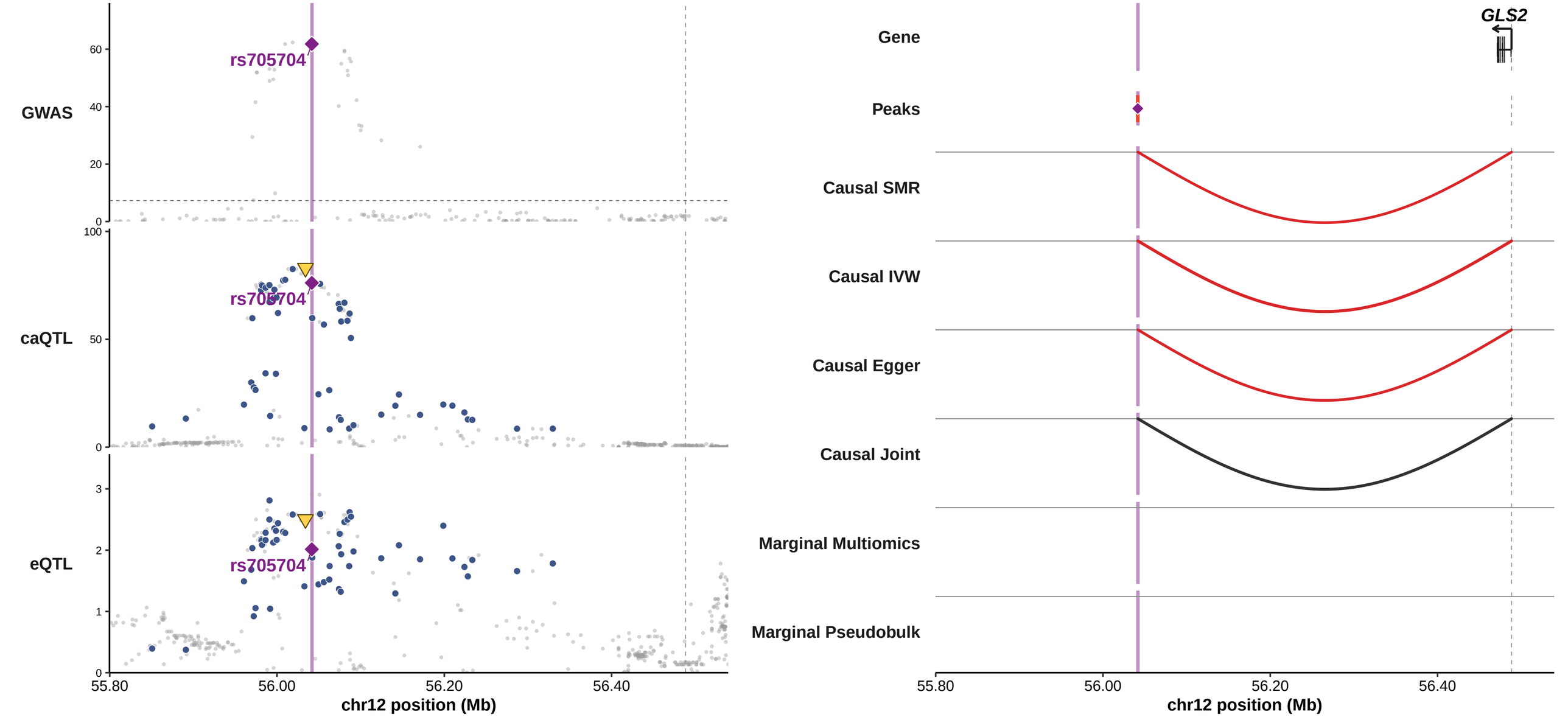
CD8_Tn_CCR7 − SYT11 − chr1:155227359−155227860 − IBD

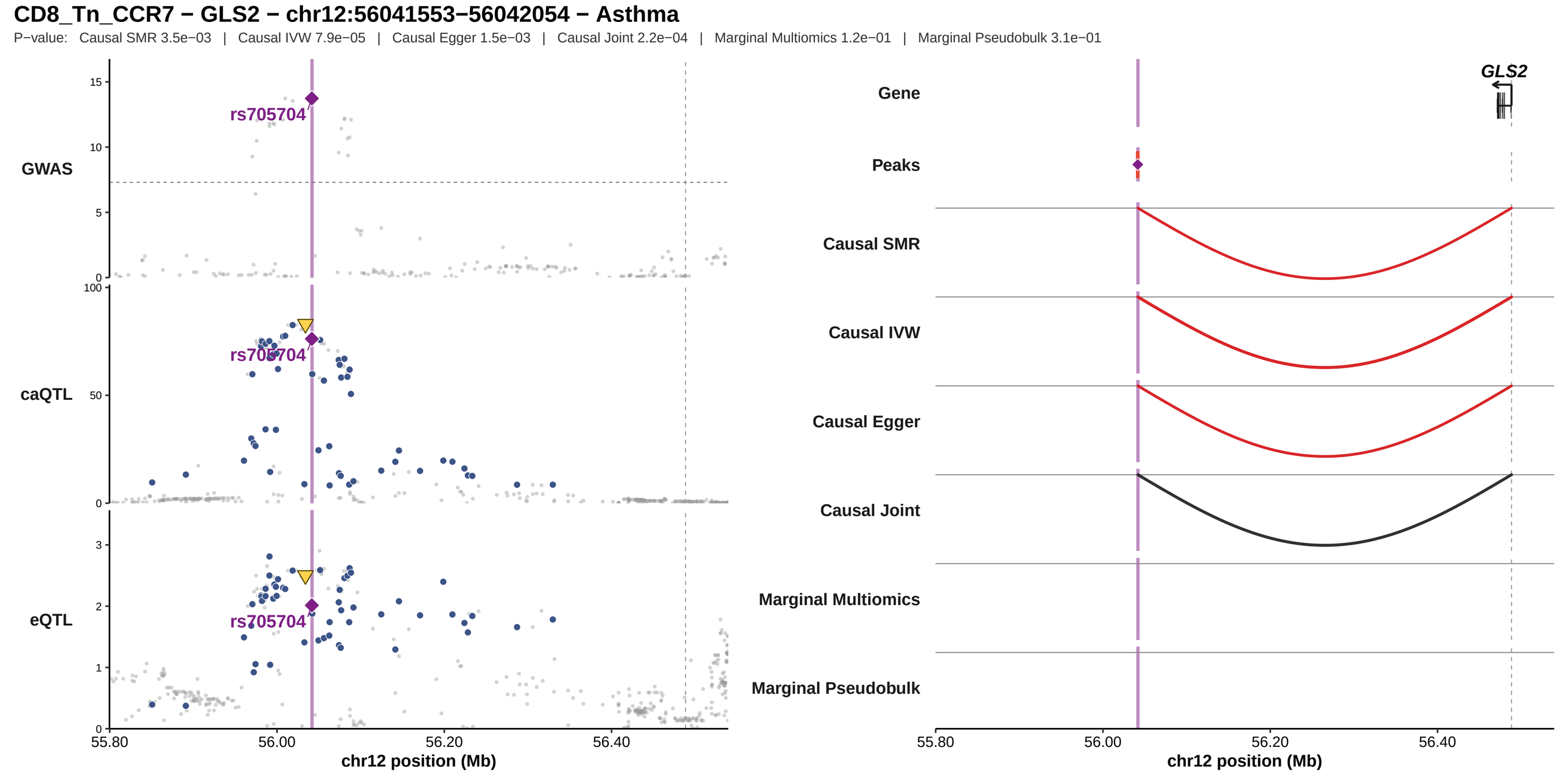
P−value: Causal SMR 6.5e−02 | Causal IVW 3.5e−05 | Causal Egger 1.5e−04 | Causal Joint 8.6e−05 | Marginal Multiomics 8.1e−01 | Marginal Pseudobulk 2.7e−01



CD8_Tn_CCR7 − GLS2 − chr12:56041553−56042054 − T1D

P−value: Causal SMR 3.5e−03 | Causal IVW 7.9e−05 | Causal Egger 1.5e−03 | Causal Joint 2.2e−04 | Marginal Multiomics 1.2e−01 | Marginal Pseudobulk 3.1e−01




CD8_Tn_CCR7 − GLS2 − chr12:56041553−56042054 − Asthma
P−value: Causal SMR 3.5e−03 | Causal IVW 7.9e−05 | Causal Egger 1.5e−03 | Causal Joint 2.2e−04 | Marginal Multiomics 1.2e−01 | Marginal Pseudobulk 3.1e−01
GWAS
caQTL
eQTL
rs705704
rs705704
rs705704
chr12 position (Mb)
55.80
56.00
56.20
56.40
Gene
GLS2
Peaks
Causal SMR
Causal IVW
Causal Egger
Causal Joint
Marginal Multiomics
Marginal Pseudobulk
chr12 position (Mb)

CD8_Tn_CCR7 − UBQLN4 − chr1:155227359−155227860 − IBD

P−value: Causal SMR 7.5e−01 | Causal IVW 7.4e−04 | Causal Egger 1.0e−04 | Causal Joint 2.7e−04 | Marginal Multiomics 2.0e−01 | Marginal Pseudobulk 3.4e−01

